A polynomial-depth learning bound for NNBayes on robust networks

Dmitry Vaintrob Kaarel Hänni OpenAI Codex

18 October 2026

Abstract

Can Bayesian neural networks learn small circuits with sample complexity polynomial in circuit width and depth, independently of the learner's width? For mean-field and stronger Gaussian prior scalings, this remains open in general. We prove such a bound after restricting the prior to networks whose predictions are stable under a calibrated ball of simultaneous internal perturbations. The networks are fully connected, with tanh activations and a linear readout. For a Boolean target given by a layered circuit of width C and depth H, one polynomial bounds the required number of examples jointly in input dimension, C, H, robustness calibration, and statistical parameters. The guarantee holds simultaneously for learners with H hidden layers at every width $N \geq C$ and every Gaussian scale $0 < t \leq 1$, including the finite-variance mean-field scale and stronger priors. Training requires exact label agreement with a fixed margin; the conclusion controls posterior test error at a smaller margin and hence sign error. The proof combines a robust implementation of each Boolean circuit with enough Gaussian probability around it, posterior compression, and stability of clipped predictions. We give the argument in full and separately assess when trained networks satisfy the prior restriction.

1 Introduction

A function with a small circuit should be learnable from relatively few examples, even by a neural network much larger than that circuit. For Bayesian neural-network learning, or NNBayes, this asks whether conditioning a prior on the training labels produces a posterior that predicts the target well. A particularly useful guarantee would have sample complexity polynomial in input dimension, target circuit width, and depth, with no dependence on the learner's potentially much larger width.

The depth requirement is substantial. A bound polynomial in target width at each fixed depth can still become exponential as depth grows. General compression arguments for Gaussian neural-network priors incur such depth costs because small errors introduced inside the network may be amplified through successive layers. The full joint polynomial learning guarantee remains open for Gaussian priors at mean-field and stronger-prior scalings without an additional internal-stability restriction. Special target classes permit better learner-depth bounds, but do not yet settle the case of arbitrary small deep circuits.

This paper obtains a joint polynomial bound by restricting the prior. We discard network parameterizations whose predictions are too sensitive to small simultaneous disturbances of their internal activations. The restriction concerns the computation performed by the network: changing an internal value slightly should not substantially alter its prediction score. We prove that every layered Boolean target circuit still has a robust implementation with enough probability under the Gaussian prior. Thus the restriction removes fragile implementations without requiring the target function to be supplied with a robust neural implementation.

The learning guarantee. Consider a fully connected tanh network with $H = L - 1$ hidden layers of width N and a real linear readout. The target is any Boolean function on d bits represented by a layered Boolean circuit of width C and at most H stages; C counts constant wires and any compilation or padding needed for this representation. Inputs are sampled independently from any fixed distribution. We start from independent Gaussian weights with layer-dependent variances, restrict the prior by the stability condition, and condition on signed training outputs greater than $1/2$. For a fixed stability calibration $c > 0$, a sufficient sample size is

$$n \geq \mathrm{poly}(d, L, C, c + c^{-1}, \varepsilon^{-1}, \log(1/\delta)).$$

With probability at least $1 - \delta$ over the sample, the posterior's average probability of a signed test output at most $1/4$ is then at most ε. The polynomial degree does not grow with depth. One sample event gives this guarantee for every $N \geq C$ and every Gaussian scale $0 < t \leq 1$ in the model of Section 2. In the exponent convention $t = N^{(1-\gamma)/2}$, this includes $\gamma = 1$ at finite width as well as every $\gamma > 1$; no width or small-variance limit is needed.

The conclusion controls sign error and posterior confidence as well as this test-margin loss. For example, the test-distribution mass of inputs at which the posterior assigns probability at most $2/3$ to the correct sign is at most 3ε. The guarantee concerns exact Bayesian conditioning; it does not supply an efficient posterior-sampling algorithm.

What the restriction requires. We clip the output at the training margin, obtaining a score in $[-1/2, 1/2]$. At every Boolean input, this score must change by at most $1/16$ under every simultaneous internal and output disturbance in a specified ball. Its radius is calibrated by depth and readout energy, as defined in Section 2. Clipping makes the condition insensitive to large changes between already confident outputs of the same sign: a change from 10 to 1 leaves the score unchanged. The condition also permits small clean outputs at unobserved inputs; the learning argument, rather than an all-input margin assumption, makes weak or incorrect test predictions rare. This is a learning theorem under a robustness restriction, not a claim that every trained network satisfies that restriction.

Why the proof gives polynomial depth. First, we implement the target circuit robustly at every width $N \geq C$ and bound the Gaussian probability of a neighborhood of that implementation. This bounds the posterior's relative entropy with respect to the original Gaussian. Second, an entropy-compression argument constructs a distribution on one finite candidate class shared by all widths and scales. At each input, the candidate and posterior computations can be coupled through small internal disturbances, with the same marginal distributions at every input. The support restriction then controls the prediction error directly, without multiplying worst-case amplification factors across layers. Finally, a uniform sampling bound for that finite class transfers training fit to test accuracy. The result is polynomial jointly in depth and the other parameters, although the polynomial is too loose to provide practical sample estimates.

Relation to prior work and verification. Compression through noise stability and all-layer margins has precedents in work of Arora et al. and Wei and Ma [1, 2]. Wei, Chen, and Ma [4] prove polynomial circuit-size sample bounds for overparameterized feedforward networks using a suitable all-layer-margin surrogate and robust circuit simulation. The role of robust simulation in avoiding exponential depth factors therefore predates this paper. Here we analyze the posterior of a specified Gaussian prior, including its target probability and a single learning guarantee uniform over widths and finite Gaussian scales. A companion Lean development verifies the qualitative main learning

theorem for the exact robust-only posterior, using a looser polynomial from a Pinsker-and-energy construction. It does not certify every quantitative constant, extension, or empirical certificate in this paper.

Organization. Section 2 specifies the prior and loss, and Section 3 states the theorem. Sections 4 and 5 give the statistical argument and robust target construction; Appendix A supplies the full compression proof. Section 6 discusses the meaning of the restriction for trained networks. The remaining appendices give certificates, weaker and localized restrictions, and the detailed empirical evidence. The main result and its proof can be read without any companion paper.

2 Model, prediction convention, and stability restriction

Let $\mathcal{X} = \{-1, 1\}^d$, let D be any distribution on $\mathcal{X}$, and let $f : \mathcal{X} \to \{-1, 1\}$ be the target. A network has $H = L - 1$ hidden layers, each of width N, tanh activations, and a linear readout:

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}) \ (2 \le s \le H), \qquad z_w(x) = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The first layer includes a bias coordinate; internal constants are carried by ordinary hidden wires. Let $\mu_{N,t}$ be the product Gaussian with entry variances t^2/d in W^1, t^2/N in the internal matrices, and t^2/N^2 in v, where $0 < t \le 1$. The exponent convention $t = N^{(1-\gamma)/2}$ therefore gives every $\gamma \ge 1$. Define

$$\kappa_N(w) = \frac{d}{N}\|W^1\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N\|v\|^2, \qquad E_v(w) = N\|v\|^2.$$

In these energy coordinates the Gaussian density is proportional to $\exp[-N\kappa_N/(2t^2)]$.

A disturbance is added after every hidden activation and at the readout:

$$h^1_\eta = \tanh(W^1 \bar{x}) + \eta^1, \qquad h^s_\eta = \tanh(W^s h^{s-1}_\eta) + \eta^s, \qquad z_w(x; \eta) = v^\top h^H_\eta + \eta^{\text{out}}.$$

The simultaneous disturbance budget is

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\text{out}}|^2. \tag{1}$$

This is an adversarial finite perturbation, not a Gaussian-noise average or a derivative evaluated only at the clean computation. Noisy hidden states need not stay in $[-1, 1]^N$.

Our training likelihood is

$$\mathcal{L}_S(w) = \prod_{i=1}^{n} \mathbf{1}\{f(X_i) z_w(X_i) > 1/2\}. \tag{2}$$

At test time we count $f(x)z_w(x) \le 1/4$ as an error. This stronger loss also bounds sign error. There is no requirement that unobserved inputs have any clean output margin.

Clip the output at the training margin:

$$T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

For a calibration parameter $c > 0$, set

$$r_c(w) = \frac{c}{\sqrt{1 + HE_v(w)}}. \tag{3}$$

Let $\mathcal{R}_{N,c}$ be the set of networks satisfying

$$|T(z_w(x;\eta)) - T(z_w(x))| \le 1/16 \quad \text{for every } x \in \mathcal{X} \text{ and every } \|\eta\|_* \le r_c(w). \tag{4}$$

The prior is exactly $\mu_{N,t}$ conditioned on $\mathcal{R}_{N,c}$. The posterior $Q_{S,N,t,c}$ additionally conditions on (2). Both conditioning events have positive probability by the target construction below. No tractable sampling algorithm is assumed or supplied by the theorem.

The clipping scale is set by the existing likelihood convention. For example, changing a raw output from 10 to 1 causes no score change. Clipping at $\pm 1/2$ gives a weaker condition than clipping at ± 1 with the same tolerance and noise radius, because $T = T \circ T_1$, where T_1 clips at ± 1, and T is 1-Lipschitz.

3 The learning theorem

A width-C, depth-H layered Boolean circuit has at most C Boolean wires at each of its H stages. We use NAND gates, copies, and negations; constant wires required by internal gate implementations are carried through the circuit and count toward C. This is a universal Boolean basis. Compiling another fixed basis into this one may change the width and depth, and those resources must be counted. Its output is one final wire. The circuit's logical depth must fit the H hidden layers; idle stages can be filled with copies.

Theorem 3.1 (Polynomial depth at every width and every $\gamma \ge 1$)**.** *Suppose f is computed by such a width-C, depth-H Boolean circuit. There is a universal polynomial P such that, for every $c > 0$ and every $0 < \varepsilon, \delta < 1$, if*

$$n \ge P(d, L, C, c + c^{-1}, \varepsilon^{-1}, \log(1/\delta)),$$

then with probability at least $1 - \delta$ over the iid labeled sample,

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \le 1/4\} \le \varepsilon \tag{5}$$

simultaneously for every $N \ge C$ and $0 < t \le 1$. The polynomial degree is independent of width, depth, and Gaussian scale. In particular, the result includes the actual finite-variance mean-field scale $\gamma = 1$ and every stronger-prior scale $\gamma > 1$.

The same theorem holds if the prior is additionally conditioned on $|z_w(x)| > 1/2$ at every input, as in the original globally margin-conditioned model. The target cylinder constructed below already satisfies this extra condition. Its probability bound and every subsequent statistical step are therefore unchanged. Under that additional condition, test-margin error coincides with sign error. Dropping the extra condition is permitted, but is not required for the result.

The width threshold concerns a layered Boolean circuit, not an arbitrary real-valued neural-network representation of the same Boolean function. The theorem does not assert that every function with a small real neural circuit has a comparably small Boolean circuit at the same depth, or that all its low-energy neural representations satisfy the stability condition. The proof instead supplies one robust implementation of each Boolean circuit, together with enough Gaussian probability around that implementation.

Define the posterior correct-margin probability $p_S(x) = Q_{S,N,t,c}\{f(x) z_w(x) > 1/4\}$. Equation (5) immediately implies

$$D\{x : p_S(x) \le 2/3\} \le 3\varepsilon, \qquad D\{x : p_S(x) \le 1/2\} \le 2\varepsilon.$$

The same bounds hold for posterior sign correctness, since the correct-sign event contains the correct-margin event. These conclusions allow arbitrary posterior mixtures; they do not require concentration on one predictor.

4 Compression and the statistical proof

The constructive compression statement needed here concerns numerical outputs before any robustness condition is used.

Lemma 4.1 (Entropy compression before the prediction comparison)**.** *For $d \geq 1$, $L \geq 2$, $K \geq 1$, $0 < r \leq 1$, and $0 < \alpha \leq 1/8$, there is a finite class $\mathcal{H}$ of real-valued bounded-weight depth-L tanh networks, independent of N, t, such that*

$$\log |\mathcal{H}| \leq \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

If $Q \ll \mu_{N,t}$ satisfies $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \leq NK/t^2$, there is one distribution Γ_Q on $\mathcal{H}$ with the following property. For every fixed x, one can couple $w \sim Q$ and $h \sim \Gamma_Q$ so that, except with probability at most α, there is a disturbance η with

$$\|\eta\|_* < r/2, \qquad |h(x) - z_w(x;\eta)| \leq r/8.$$

The couplings and disturbances may depend on x, but their two marginals are the same at every input. No robustness or clean-margin assumption is made on Q.

The proof is constructive: it retains a bounded number of input directions carrying conditional Gaussian relative entropy in each layer, replaces the remaining directions by Gaussian randomness, samples a bounded number of source neurons, and rounds a bounded-size network to a fixed grid. The resulting virtual computation is the original network under small postactivation disturbances. The appendix gives the complete construction; the prediction comparison below is the first step that uses stability.

Lemma 4.2 (Readout truncation)**.** *Suppose a posterior is the raw Gaussian conditioned on an event of mass at least $\exp(-ND_*/t^2)$, where $D_* \geq 1$. For $0 < \beta \leq 1/2$, put*

$$B = \max\{8, 8[D_* + \log(1/\beta)]\}.$$

It assigns mass at most β to $E_v > B$. Its conditional law Q' on $E_v \leq B$ satisfies $D_{\mathrm{KL}}(Q'\|\mu_{N,t}) \leq N(D_*+1)/t^2$.

Proof. Under the Gaussian, NE_v/t^2 has the chi-square distribution with N degrees of freedom. Its exponential moment gives, for $B \geq 8$,

$$\mu_{N,t}\{E_v > B\} \leq e^{-NB/(4t^2)} 2^{N/2} \leq e^{-NB/(8t^2)}.$$

Divide by the conditioning-event probability and use $N/t^2 \geq 1$. The further conditioning event has raw probability at least $(1-\beta)e^{-ND_*/t^2}$, so its negative logarithm is at most $ND_*/t^2 + \log[1/(1-\beta)] \leq N(D_*+1)/t^2$. □

Statistical proof of Theorem 3.1. The next section constructs a subset of the support with correct noisy margin on every input and raw mass at least e^{-ND_*/t^2}, where $D_* = \mathrm{poly}(d, L, C, c)$ is independent of N, t. It is contained in the posterior conditioning event for every correctly labeled sample. Apply Lemma 4.2 with $\beta = \varepsilon/4$, and write Q' for the retained law. Set

$$K = D_* + 1, \qquad r = \min\{1/6, c/\sqrt{1+HB}\}, \qquad \alpha = \varepsilon/12.$$

Every retained network satisfies the score-stability condition at the common radius r. Lemma 4.1 and the 1-Lipschitz property of T therefore give pointwise couplings with

$$\Pr\{|T(h(x)) - T(z_w(x))| > 1/12\} \le \alpha,$$

because $1/16 + r/8 \le 1/12$.

Use the bounded ramp loss

$$\psi(u) = \begin{cases} 1, & u \le 1/3, \\ 5 - 12u, & 1/3 < u < 5/12, \\ 0, & u \ge 5/12. \end{cases}$$

At a training input, $f(x)z_w(x) > 1/2$ implies $f(x)T(z_w(x)) = 1/2$. A successful coupling therefore gives $f(x)T(h(x)) \ge 5/12$, so its ramp loss is zero. The candidate's averaged empirical ramp risk is at most α. At a test input with $f(x)z_w(x) \le 1/4$, a successful coupling gives $f(x)T(h(x)) \le 1/3$, so its ramp loss is one.

For any finite class of $[0,1]$-valued losses, a relative Chernoff bound and a union bound give, with probability at least $1 - \delta$,

$$\mathbb{E}_D \psi(f(X)T(h(X))) \le \frac{2}{n} \sum_{i=1}^{n} \psi(f(X_i)T(h(X_i))) + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n} \quad (h \in \mathcal{H}).$$

For completeness, $e^{-u} \le 1 - (1 - e^{-1})u$ on $[0,1]$ gives the usual lower-tail exponent at least the mean divided by eight. This proves the displayed bound for one loss, and a union bound proves simultaneity. Average under the sample-dependent law $\Gamma_{Q'}$, apply the population comparison, and restore the discarded posterior mass. The result is

$$\Pr_{Q_{S,N,t,c},D}\{f(X)z_w(X) \le 1/4\} \le \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(1/\delta)]}{n}.$$

Choose n so the last term is at most $\varepsilon/2$. Every quantity entering $\log|\mathcal{H}|$ is bounded by a universal polynomial in the theorem's parameters. Its class and the displayed sample event are independent of N, t, establishing the simultaneous statement. □

5 A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \ge C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \le N/k_a \le 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16\log 9 + 8\log(4L)},$$
$$A = \max\{32b_L, 2\log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\,W^1$, $\sqrt{N}\,W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A $1/4$-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least $1/2$.

Finite disturbance robustness. Allow a separate RMS disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \le \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \le e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \le 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \ge A/2 \quad \Longrightarrow \quad |\tanh(q+u) - \epsilon| \le 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h^{s-1}_\eta\|_2/\sqrt{N} \le 1 + \tau_0$, it follows that

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \le A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x)z_w(x;\eta) \ge \frac{15a}{16} - \rho(1+\tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \le \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \ge a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v - v_*\| \le \rho$. Therefore $E_v \ge a^2 C/8$, whence

$$r_c(w) \le \frac{2\sqrt{2}c}{a\sqrt{C}} \le \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \le 1/4$, and any center m,

$$\Pr\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \ge \exp(-ND_*/t^2), \qquad D_* = B_0 + q \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{6}$$

Here $D_* \ge 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \ge C$.

6 What the restriction means for trained networks

The theorem's prior restriction is motivated by stability of a computation, but its empirical relevance is a separate question. It quantifies over every Boolean input and every allowed finite disturbance. Stability to random noise on a test set is useful evidence, but has different quantifiers. Published experiments report attenuation of injected activation noise in a VGG-19 trained on CIFAR-10 [1], and improved noise stability in residual networks relative to matched plain networks [5]. They do not establish the support condition used here.

There are concrete sufficient conditions. If every internal hidden-layer map is nonexpansive, meaning it does not increase Euclidean distances, then the readout-and-depth calibration makes the required bound hold for $c \le 1/16$. Individual layers need not all be nonexpansive: bounds on their combined effects, or saturation of tanh along the reachable computations, can also certify stability. Appendix B develops these tests. A calibration c that decreases only polynomially with depth still gives a polynomial sample bound.

Our experiments and exact-arithmetic certificates give mixed evidence. Some image classifiers fitted to every training label satisfy a global stability certificate at the stated small calibrations, and two trained 64-hidden-layer Boolean classifiers satisfy the condition on their entire finite input domains. Another fitted image model has a verified confident-input counterexample even though it satisfies certificates on all 10,000 official held-out images at the same radius. The experiments, domain conventions, and certificate scopes are reported in Appendices M, N, and O. In particular, a certificate at one input covers that input only, and failure to find an adversarial disturbance is not a certificate.

Gradient descent does not automatically enforce the condition: Appendix K gives an exact construction that is stable on the data distribution and fragile elsewhere. Localized versions of the prior can allow exceptional input regions when their complexity is controlled (Appendix D). Whether the global condition, or such a localized version, commonly holds for relevant trained networks remains a substantive empirical question. These investigations concern membership in the restricted prior; they do not identify the distribution of gradient-descent outputs with the Bayesian posterior analyzed by the theorem.

A Finite posterior compression by internal-noise coupling

This appendix gives the compression construction used in the main theorem. It tracks relative entropy only in the original weights that have not yet been processed. The compressed prefix need not have a Gaussian density. No stability assumption is needed until the coupling is used in the main statistical argument.

Theorem A.1 (Finite compression before invoking stability)**.** *Fix a scalar 1-Lipschitz activation $|\phi| \le 1$, and parameters $d \ge 1$, $L \ge 2$, $K \ge 1$, $0 < r \le 1$, and $0 < \alpha \le 1/8$. There is a finite class $\mathcal{H} = \mathcal{H}(d, L, K, r, \alpha)$ of real-valued bounded-weight depth-L network outputs with*

$$\log |\mathcal{H}| \le \mathrm{poly}(d, L, K, r^{-1}, \alpha^{-1}).$$

The polynomial and class are independent of width N and scale $0 < t \le 1$. For every $Q \ll \mu_{N,t}$ with $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$, there is a law Γ_Q on $\mathcal{H}$ such that, at every Boolean input x, one can couple $w \sim Q$, $h \sim \Gamma_Q$, and a disturbance η so that

$$\mathbb{P}\{\|\eta\|_* < r/2, \quad |h(x) - z_w(x;\eta)| \le r/8\} \ge 1 - \alpha.$$

The coupling may depend on x; its two specified marginals do not.

We give the proof, including deliberately loose polynomial choices. Two elementary entropy facts organize the construction. First, for a standard D-dimensional Gaussian G, the entropy variational inequality gives

$$\mathbb{E}_P\|Z\|^2 \le 4D_{\mathrm{KL}}(P\|G) + 2D\log 2. \tag{7}$$

Second, Gaussian Talagrand [3] gives $W_2^2(P, G) \le 2D_{\mathrm{KL}}(P\|G)$.

Lemma A.2 (Entropy of the unprocessed weights)**.** *Let A, X, Y be random variables, with fixed reference laws P_X, P_Y, and put*

$$B = \mathbb{E}_A D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y).$$

Let J be a finite message, possibly depending on (A, X, Y), and let $H = f(A, J, X)$. If A' is a randomized function of (A, J, H) whose fresh randomness is conditionally independent of Y, then

$$\mathbb{E}_{A'} D_{\mathrm{KL}}(Q_{Y|A'}\|P_Y) \le B + H(J \mid A).$$

Proof. The left side equals $D_{\mathrm{KL}}(Q_Y\|P_Y) + I(Y; A')$. Data processing and

$$I(Y; J, H \mid A) \le I(Y; X, J \mid A) \le I(Y; X \mid A) + H(J \mid A)$$

bound it by $\mathbb{E}_{A,X} D_{\mathrm{KL}}(Q_{Y|A,X}\|P_Y) + H(J \mid A)$, which is at most the claimed bound by the KL chain rule. These expressions can equivalently be justified by finite partitions and lower semicontinuity. □

Here A will be a constructed prefix, X the current original weight block, and Y all later original blocks. The lemma imposes no reference law on A.

Lemma A.3 (Whole-layer active directions)**.** *Condition on (A, Y) and let X be a standardized Gaussian-reference matrix, with conditional relative entropy $R(A, Y)$. For vectors $b_x(A)$ of norm at most one, one can select a list J of inputs and its span $U = \mathrm{span}\{b_x : x \in J\}$ with*

$$\dim U \le \frac{t^2 R(A, Y)}{N\delta}$$

such that preserving $H = tXU$ and replacing the orthogonal coordinates by fresh Gaussian coordinates changes the physical evaluation vectors tXb_x, under a suitable coupling, by expected squared RMS at most 2δ at each x. The candidate matrix law preserves (A, Y, H) and does not depend on x.

Proof. Use a fixed ordering of the finite input set. Whenever the averaged conditional KL of the entire residual evaluation vector Xv exceeds $N\delta/t^2$, add the unit residual direction v of that input. Each query is selected from the conditional law given (A, Y), averaging over active-coordinate values; thus J is a function of (A, Y). Orthogonal Gaussian coordinates are independent, so the active-coordinate entropy increases by precisely this conditional KL. The displayed rank bound follows. At termination, Gaussian Talagrand on the entire evaluation vector, multiplication by the squared residual-vector length, and division by N/t^2 give the error bound. Scalar-vector couplings extend to full matrices by disintegration. The basis of U is a measurable function of A, J under fixed Gram–Schmidt; it has no further dependence on Y once those variables are supplied. □

Parameter choices. Write

$$\begin{aligned} e &= \alpha r^2/(1000L), & \eta &= \alpha/(100L), & q &= \lceil 2K/(e\eta)\rceil, \\ D_0 &= [(d+1)q+2]\log 2, & V &= 8K+2q, & B_H^2 &= 4V/e, \\ T &= 128(q+1)(B_H^2+1)/e, & a &= e/[128(B_H^2+1)], & & \\ K_0 &= (4K+2\log 2)/\eta, & B_v &= 16K_0/r. & & \end{aligned}$$

Choose m to be the least positive integer no smaller than both

$$\begin{aligned} &10000T^4(q+1)^2a^{-2}[d+\log(10(q+1)/\eta)+1], \\ &10000B_v^2r^{-2}[d+\log(10/\eta)+1], \end{aligned}$$

and put

$$N_0 = \max\{m, \lceil LD_0/K\rceil, 1\}, \quad B_G = \max\{2, 2\sqrt{\log(3m/e)}\}, \quad B_1^2 = 4[4K+2(d+1)\log 2]/e.$$

All quantities have a fixed polynomial upper bound in the theorem parameters.

The small-width case. If $N \le N_0$, let $P_0 = N_0(d+1) + (L-2)N_0^2 + N_0$. Applying (7) to all standardized weights, and then multiplying by t^2, gives

$$\mathbb{E}_Q\|tZ_{\mathrm{all}}\|^2 \le 4N_0K + 2P_0\log 2.$$

With probability at least $1-\alpha/4$, all actual weights are bounded by

$$B_{\mathrm{small}} = [4(4N_0K + 2P_0\log 2)/\alpha]^{1/2}.$$

Keep the original network on this event and use a default zero-readout network otherwise. Padding with nodes having zero outgoing weights preserves the original output. It remains to treat $N > N_0$.

The virtual prefix and its entropy budget. The constructed prefix A has N virtual nodes in its current layer. Each virtual row acts on m stored nodes of the preceding layer. The auxiliary description may keep all N rows; the final network will keep just the rows selected at the following stage. Track the conditional KL of the unprocessed *original* weights. By Lemma A.2, it is at most

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2. \qquad (*)$$

This is not a KL bound for the singular compressed prefix itself.

For the first layer, zero every physical standardized row $tZ_{1,i}$ of norm greater than B_1, and retain all other rows. Bounded activations and (7) bound the expected squared RMS activation change at each input by e. This deterministic prefix channel does not increase the future entropy budget.

An internal layer: retain active data. Condition on the current prefix A and the future original weights Y. Apply Lemma A.3 to $b_x = h_A(x)/\sqrt{N}$ and $\delta = e$. Abort if more than q inputs are needed. By (*) and Markov, this failure has probability at most η. The successful query list, together with an abort flag, has a code of at most $[(d+1)q+2]$ bits. Its entropy is at most D_0.

Let $H = tXU$ be the physical active matrix. Averaging the Gaussian entropy moment bound, with the success indicator rather than conditioning on success, gives

$$\mathbb{E}\left[\frac{\|H\|_F^2}{N};\ \text{success}\right] \le 4t^2 B_s/N + 2t^2 q \log 2 \le V.$$

Zero every row H_i whose norm exceeds B_H. The expected squared RMS activation change from this operation is at most $4V/B_H^2 = e$. All rows are clipped before any later selection, including selections that may depend on future weights.

An internal layer: reconstruct using m source nodes. For the orthonormal active basis let $u(j) = \sqrt{N}\, U_{j,:}$; uniform source sampling gives $\mathbb{E}_j u(j)u(j)^\top = I$. Clip the *whole vector* at radius T: $c(j) = u(j)\mathbf{1}_{\{\|u(j)\| \le T\}}$. Then

$$\mathbb{E}cc^\top \preceq I, \qquad \mathbb{E}\|u - c\| \le q/T.$$

Sample m uniform source indices S_j. Entrywise Hoeffding and a union bound give, with probability at least $1-\eta$,

$$\|\widehat{G} - \mathbb{E}cc^\top\|_{\mathrm{op}} \le a, \quad \sup_x \|\widehat{v}(x) - \mathbb{E}ch_A(x)\| \le a,$$
$$\sup_{x,y} |\widehat{K}(x,y) - K(x,y)| \le a,$$

where $\widehat{G} = m^{-1}\sum_j c(S_j)c(S_j)^\top$, $\widehat{v}(x) = m^{-1}\sum_j c(S_j)h_{A,S_j}(x)$, and $\widehat{K}$ is the empirical feature covariance; its population counterpart is $K(x,y) = N^{-1}\sum_i h_{A,i}(x)h_{A,i}(y)$. Abort on failure; this event depends only on A, J, S, so it does not reveal further future information.

Set $\widetilde{c}_j = c(S_j)/\sqrt{1+a}$, and let D have rows $\widetilde{c}_j^\top/\sqrt{m}$. Then $D^\top D \preceq I$ and $I - DD^\top \succeq 0$. Independently for every target virtual row draw $\xi_i \sim N(0, I - DD^\top)$ and use coefficients

$$\widehat{w}_{ij} = m^{-1} H_i^{\mathrm{clip}} \cdot \widetilde{c}_j + t\xi_{ij}/\sqrt{m}. \tag{8}$$

These are actual neural-network rows on the m sampled nodes. Their active feature vector is $\widehat{\psi}(x) = m^{-1}\sum_j \widetilde{c}_j h_{A,S_j}(x)$, and the lazy covariance is exactly $t^2(\widehat{K} - \widehat{\psi}^\top\widehat{\psi})$. For the ideal field, write $\psi(x) = U^\top b_x$ and $C(x,y) = K(x,y) - \psi(x)^\top\psi(y)$. Direct calculation gives

$$\sup_x \|\widehat{\psi}(x) - \psi(x)\| \le q/T + 2a =: \zeta, \qquad \sup_{x,y} |\widehat{C}(x,y) - C(x,y)| \le a + 2\zeta.$$

Here $\widehat{C} = \widehat{K} - \widehat{\psi}^\top\widehat{\psi}$. Bessel's inequality and $D^\top D \preceq I$ give $\|\psi(x)\|, \|\widehat{\psi}(x)\| \le 1$, which justify the covariance bound. At a fixed input, coupling scalar Gaussian evaluations with a common standard normal bounds the squared preactivation transport error, per row, by

$$B_H^2\zeta^2 + t^2(a + 2\zeta) \le e/8.$$

Extend these scalar couplings to full row fields. The candidate weight law is the same at every input; there is no claim of a single Gaussian-field coupling that is close at all inputs simultaneously.

Finally clip *all* coordinates ξ_{ij} to $[-B_G, B_G]$ before later row selection. Since each has variance at most one,

$$\mathbb{E}\left|\frac{t}{\sqrt{m}}\sum_j(\xi_{ij}-\operatorname{clip}_{B_G}\xi_{ij})h_{A,S_j}(x)\right|^2 \le 3me^{-B_G^2/4} \le e.$$

Thus every virtual coefficient is bounded by $B_H T/m + B_G/\sqrt{m}$, without a union bound over N.

All newly constructed data are a randomized function of A, J, H. Lemma A.2 therefore proves (*) for the next step. Combining entropy replacement, active clipping, finite reconstruction, and Gaussian clipping by $(a_1+\cdots+a_4)^2 \le 4\sum_i a_i^2$, the expected squared RMS activation injection at this layer is less than $20e$.

The readout and final network. The original readout marginal has never changed. Its physical moment satisfies $\mathbb{E}_Q[N\|v\|^2] \le 4K + 2\log 2$. Abort if $N\|v\|^2 > K_0$, an event of probability at most η. Zero coefficients with $|Nv_i| > B_v$. This changes the output uniformly on bounded virtual feature vectors by at most $K_0/B_v = r/16$. Sampling m uniform last-layer sources and using readout coefficients $(Nv_i\mathbf{1}_{|Nv_i|\le B_v})/m$ adds at most $r/16$ uniformly on $\{-1,1\}^d$, except with probability η, by Hoeffding and the chosen m.

At each stage, retain in the final network just the m source rows selected from the preceding virtual layer. Each such row already acts on the previously stored m nodes. The result is an ordinary width-m network; repeated indices are duplicate nodes. Its weights are bounded by

$$B_{\text{large}} = \max\{B_1, B_H T/m + B_G/\sqrt{m}, B_v/m, 1\}.$$

The larger virtual construction is not part of the final predictor.

The original network receives all perturbations. Fix an input x and glue the conditional couplings, retaining the original weight history. The current and future original blocks are untouched until their own stages. Define

$$\eta^s = h^s_{\text{virtual}}(x) - \phi(W^s_{\text{original}} h^{s-1}_{\text{virtual}}(x)).$$

The analogous readout difference is η^L. Later stages leave earlier computational prefixes unchanged, so the final candidate output is exactly the *original* network's output under these postactivation injections. No intermediate robustness assumption is made.

For precision, the canonical construction draws its new Gaussian coefficients independently of the discarded current block given (A, J, H). The comparison coupling may correlate them with that discarded block. It preserves the same joint marginal of the new prefix and the future original blocks, as well as the original full-network marginal, but need not preserve the canonical joint law with all discarded weights. The next entropy step uses the first marginal; original-network robustness uses the second. There is no conflicting demand for independence in the comparison coupling itself.

More explicitly, the induction invariant is that the comparison law has full original-weight marginal Q and the canonical marginal of (A, X, Y). The conditional transports supply a joint law of (A, X, Y, A') with this incoming marginal and the prescribed outgoing (A', Y) marginal. Extend it by the previous conditional law of the past original weights given (A, X, Y). The extension keeps the complete original weight law unchanged, preserves all earlier computational prefixes, and establishes the same invariant for the next layer. This is ordinary disintegration on finite-dimensional standard Borel spaces.

Abort probability is at most $2L\eta$. The unconditional moment bound gives

$$\mathbb{P}\left\{\text{no abort and } \sum_{s<L} \|\eta^s\|^2/N > r^2/8\right\} \le 160Le/r^2 = 0.16\alpha.$$

Otherwise the readout injection is at most $r/8$, and total squared noise is at most $9r^2/64 < r^2/4$. The unrounded candidate is therefore exactly the original network evaluated with a disturbance of norm less than $r/2$. The exceptional probability is less than $\alpha/4$.

A fixed finite class. Pad candidates to width $M = N_0$ and take $B = \max(B_{\text{small}}, B_{\text{large}}, 1)$. Round every parameter to a fixed grid on $[-B, B]$ of mesh at most

$$h = \frac{r}{16(d+L+1)(2M(B+1))^L}.$$

The bounded-activation, Lipschitz layer recursion gives uniform output change at most $r/8$. Thus rounding gives the stated numerical comparison at every successfully coupled input. Writing $P = M(d+1) + (L-2)M^2 + M$, the grid class satisfies

$$\log|\mathcal{H}| \le P \log\left[3 + \frac{32B(d+L+1)(2M(B+1))^L}{r}\right].$$

Although grid precision is exponentially small in L, its number of bits is polynomial in L. All other bounds above are polynomial too. Rounding is a fixed map on candidate networks, giving the common marginal Γ_Q and proving Theorem A.1.

B Sufficient stability conditions and finite-radius certificates

The support condition is a finite-perturbation requirement. Clean Jacobians, loss Hessians, or held-out margins alone do not prove it. The following certificates make the condition explicit. They are sufficient tests for membership in the support already defined, not additional restrictions on that support. Restricting the prior to only networks passing a particular conservative test would require a new target-mass argument.

B.1 Global spectral and diagonal-metric conditions

For $1 \le j \le H$, put

$$p_j = \prod_{s=j+1}^{H} \|W^s\|_{\text{op}}, \qquad S = \sum_{j=1}^{H} p_j^2,$$

where the empty product is one. For every input and every finite postactivation disturbance,

$$|z_w(x;\eta) - z_w(x)| \le \sqrt{1 + E_v S}\, \|\eta\|_*. \tag{9}$$

Indeed, tanh is 1-Lipschitz, so the final hidden-state difference is at most $\sum_j p_j \|\eta_j\|_2$ in Euclidean norm. Apply the readout, add the output disturbance, and use Cauchy–Schwarz in the normalized noise coordinates. In particular,

$$c^2 \frac{1 + E_v S}{1 + H E_v} \le \frac{1}{256} \tag{10}$$

is sufficient for the clipped support condition globally. If every internal operator norm is at most one, then $S \le H$; hence every such network qualifies whenever $c \le 1/16$, without a restriction on its first layer or readout.

The suffix products permit compensating expansion and contraction. For scalar internal weights alternating between 2 and $1/2$, every $p_j \le 2$, so $S \le 4H$. Thus $c \le 1/32$ suffices, even though a positive fraction of the layers expand by a factor two. More generally, if $p_j \le R$ for every j, where $R \ge 1$, then the output change at the calibrated radius is at most cR. The point of controlling suffix products is that noise can enter at every depth: a bound on only the product of all layer norms does not control this situation.

A sharper global certificate can ignore expansive directions that never affect the output. Suppose diagonal positive semidefinite matrices $M_1, \dots, M_H$ satisfy

$$M_H \succeq vv^\top, \qquad (W^s)^\top M_s W^s \preceq M_{s-1} \quad (2 \le s \le H). \tag{11}$$

Then, writing $Q_M = N \sum_s \lambda_{\max}(M_s)$,

$$|z_w(x;\eta) - z_w(x)| \le \sqrt{1 + Q_M}\, \|\eta\|_*. \tag{12}$$

To prove this, use the diagonal seminorms $\|u\|_{M_s} = \sqrt{u^\top M_s u}$. Coordinatewise 1-Lipschitzness implies

$$\|\Delta h_s\|_{M_s} \le \|\Delta h_{s-1}\|_{M_{s-1}} + \|\eta_s\|_{M_s}.$$

At the first layer $\Delta h_1 = \eta_1$. Sum, apply $M_H \succeq vv^\top$, and use Cauchy–Schwarz. Diagonality matters: a coordinatewise contraction need not contract an arbitrary quadratic form with off-diagonal entries.

Taking $M_s = \|v\|^2 p_s^2 I$ recovers the spectral certificate. With $W^s = \mathrm{diag}(1, 10)$ and $v = (1, 0)^\top$, however, taking $M_s = \mathrm{diag}(1, 0)$ gives $Q_M = HE_v$ despite the exponentially large spectral suffix products. For fixed weights, minimizing $\sum_s b_s$ under (11) and $0 \preceq M_s \preceq b_s I$ is a semidefinite program. Penalizing this certificate during training is a concrete proposed regularizer; it is not a claim that ordinary gradient descent automatically produces such matrices with a favorable value.

B.2 A clean-Jacobian bound with an explicit finite-radius remainder

There is also a sufficient condition that uses the clean trajectory but controls the nonlinear remainder quantitatively. Fix x and write $a_s = W^s h_{s-1}$ and $D_s = \mathrm{diag}(\mathrm{sech}^2 a_s)$ for $s \ge 2$. On the stacked space $\mathbb{R}^{HN}$, let A be strictly block lower bidiagonal, with block $A_{s,s-1} = D_s W^s$ and all other blocks zero. Define

$$K = (I - A)^{-1}, \qquad k = \|K\|_{\mathrm{op}}, \qquad \ell = v^\top P_H K, \qquad g = \|\ell\|_2,$$

where P_H selects the last hidden-state block. The inverse exists because A is nilpotent. Put

$$b = \max_{2 \le s \le H} \|W^s\|_{\mathrm{op}}, \qquad b_\infty = \max_{\substack{2 \le s \le H \\ 1 \le i \le N}} \|(W^s)_{i,:}\|_2.$$

When $H = 1$, set $b = b_\infty = 0$. Given $r > 0$, set

$$R = r\sqrt{N}, \qquad \theta = 2bb_\infty k^2 R.$$

Proposition B.1 (Finite-radius Jacobian certificate)**.** *If $\theta < 1$, then every disturbance with $\|\eta\|_* \le r$ satisfies*

$$|z_w(x;\eta) - z_w(x)| \le r\sqrt{1 + Ng^2(1+\theta)^2}. \tag{13}$$

Consequently a right-hand side at most $1/16$ certifies the support condition at x and radius r.

Proof. Let X stack all hidden-state differences, and let η_h stack the hidden injections. Taylor expansion about the clean preactivations gives

$$X = AX + \eta_h + e(X), \qquad X = K(\eta_h + e(X)).$$

Let B be the block operator sending X to the stacked internal preactivation changes. Then $\|B\|_{\mathrm{op}} \le b$ and $\|BX\|_\infty \le b_\infty \|X\|_2$. Since $\sup_{u\in\mathbb{R}} |\tanh''(u)| = 4/(3\sqrt{3}) < 1$,

$$\|e(X)\|_2 \le \tfrac{1}{2}\|BX\|_\infty \|BX\|_2 \le \tfrac{1}{2} b b_\infty \|X\|_2^2. \tag{14}$$

Put $u = \|\eta_h\|_2 \le R$. If $u = 0$, then $X = 0$. If $u > 0$, vary the hidden injection continuously from zero to η_h. The corresponding feedforward states depend continuously on this variation. At a first crossing of $\|X\|_2 = 2ku$, the preceding identity would imply

$$2ku \le ku + \tfrac{1}{2} k b b_\infty (2ku)^2 = ku\big(1 + 2 b b_\infty k^2 u\big) < 2ku,$$

a contradiction. Thus $\|X\|_2 < 2ku$ throughout the homotopy. Applying the last-layer readout to the exact identity gives

$$|v^\top P_H X| \le gu + \tfrac{1}{2} g b b_\infty (2ku)^2 \le g(1+\theta)u.$$

Finally combine this with $|\eta_{\mathrm{out}}|$ by Cauchy–Schwarz, using $u^2/N + |\eta_{\mathrm{out}}|^2 \le r^2$. □

The proposition permits cancellations in the clean transport and saturation along the clean trajectory. Its curvature condition is essential: a small clean Jacobian alone gives no positive finite-radius guarantee. One can upper-bound k using the $H \times H$ block-norm matrix with entries

$$T_{sj} = \begin{cases} 1, & s = j, \\ \|D_s W^s \cdots D_{j+1} W^{j+1}\|_{\mathrm{op}}, & s > j, \\ 0, & s < j. \end{cases}$$

Then $k \le \|T\|_{\mathrm{op}}$; replacing each product norm by the product of its factor norms is another valid, more conservative upper bound. The quantity g^2 is the sum of squared Euclidean norms of the clean readout derivatives with respect to all hidden injections.

Finite reachable-tube certificates offer a different refinement: they bound derivatives throughout explicitly bounded reachable activation intervals and propagate those bounds. They can succeed when the preceding single global curvature budget fails. Conversely, retaining signed products in K may exploit cancellation lost by scalar interval propagation. Neither method generally dominates the other. Both certify finite disturbances; an uncorrected clean-Jacobian test does not.

B.3 Using clipping and controlling the domain

Any raw-output certificate $|z_w(x;\eta) - z_w(x)| \le B(x)$ can be converted to the sharper clipped certificate

$$\max\{T(z_w(x) + B(x)) - T(z_w(x)),\, T(z_w(x)) - T(z_w(x) - B(x))\} \le 1/16.$$

Monotonicity of clipping proves this implication. In particular, large raw-output variation can be harmless when the entire certified interval lies beyond the same training-margin endpoint.

Global spectral or diagonal-metric certificates hold on the entire input domain automatically. The trajectory-dependent certificates are pointwise. Testing them at independent held-out inputs

estimates their success probability for one fixed network; it does not establish global membership or the low-complexity exceptional-set hypothesis of the localized posterior theorem. To certify an input region, its clean activation ranges and noise-perturbed ranges must both be controlled uniformly over that region. The distinction is necessary even for a finite pixel grid or a Boolean encoding of images.

C Variants of the prediction-stability condition

C.1 The tolerance is determined by the margin gap

The constants in the main condition arise from the gap between the training and test margins. They need not be treated as separate modeling choices.

Corollary C.1 (General training and test margins)**.** *Fix $m > 0$ and $0 \le u < m$, and put $\Delta = m - u$. Replace the training likelihood by $f(X_i)z_w(X_i) > m$, the test error by $f(X)z_w(X) \le u$, and the clipping map by $T_m(z) = \max\{-m, \min\{m, z\}\}$. Require clipped-score variation at most $\Delta/4$ on the calibrated ball of radius $c/\sqrt{1 + HE_v}$. Then the main theorem holds with a sample bound polynomial also in m and Δ^{-1}.*

Proof. Use readout coefficient $a = \max\{2, 2m, 256c\}$ in the target cylinder. Its noisy signed output is greater than m, its calibrated radius is still at most τ_0, and its Gaussian negative log-mass remains polynomial in m, c, d, L, C. After the usual readout truncation use

$$r = \min\{1, 2\Delta/3, c/\sqrt{1 + HB}\}.$$

Successful compression changes the clipped score by at most $\Delta/4 + r/8 \le \Delta/3$. Training scores are therefore at least $m - \Delta/3$, whereas a test margin failure has candidate score at most $u + \Delta/3$. Use a $[0, 1]$-valued ramp which is zero above the former threshold and one below the latter; these thresholds are separated by $\Delta/3$. The same relative-Chernoff argument proves the result. □

The main theorem is $m = 1/2$, $u = 1/4$, so $\Delta/4 = 1/16$. If only the sign is scored at test time, take $u = 0$: the permissible score-variation tolerance increases to $1/8$. This concerns the raw-output test margin, separately from whether posterior confidence is judged at $1/2$ or $2/3$.

C.2 Disturbances confined to valid activation states

One may restrict the support condition to disturbances whose complete noisy hidden trajectory lies in $[-1, 1]^N$ at every layer. This permits perturbations only to values within the range of the tanh activation. The same main theorem holds. Indeed, before grid rounding, every successful virtual hidden state in the constructive entropy compressor is a tanh output. Thus the disturbances used in its prediction comparison satisfy this extra state restriction. The separate output injection and subsequent grid-rounding error are treated exactly as before. The original target cylinder was stable even without the state restriction and therefore retains its probability lower bound.

This condition is weaker than unrestricted additive-disturbance stability. A negative empirical attack for this variant must verify the state constraint at every intermediate layer; an attack outside the activation range does not refute it. The primary empirical condition can remain the unrestricted one, with this variant stated separately.

C.3 A condition involving hidden disturbances only

The scalar output injection can also be omitted from the prior condition. For $E_v > 0$ define the hidden-only calibrated radius

$$r_c^{\mathrm{hid}}(w) = \frac{c}{\sqrt{HE_v(w)}}.$$

When $v = 0$, the output is identically zero and hidden disturbances cannot change it, so declare hidden-only score stability automatically satisfied. Require score variation at most $1/16$ for every hidden disturbance with $\sum_s \|\eta^s\|^2/N \le (r_c^{\mathrm{hid}}(w))^2$, with no output injection. The main theorem continues to hold for this Gaussian-conditioned prior.

To see this, use the same target cylinder, which has $r_c^{\mathrm{hid}} \le \tau_0$, and after readout truncation choose

$$r = \min\{1/30, c/\sqrt{HB}\}.$$

The raw compression lemma supplies hidden disturbances of norm less than $r/2$, plus an output injection of magnitude less than $r/2$. Use stability only for the hidden disturbances and treat the output injection as an approximation error. Including grid rounding, the total clipped-score error is at most

$$\frac{1}{16} + \frac{r}{2} + \frac{r}{8} \le \frac{1}{12}.$$

The training/test ramp proof is unchanged. In a nonexpansive stack the hidden-only gain bound is $\sqrt{HE_v}$, so this calibration yields raw output variation at most c without the additional scalar-noise term.

C.4 Amplification is not evidence about unmodified trained weights

Every finite-cube network with no zero clean outputs eventually satisfies the calibrated clipped condition after a sufficiently large positive readout amplification. To verify this, continuity and finiteness of the input cube give a hidden-disturbance radius $\rho > 0$ and $b > 0$ such that the unamplified noisy output has its clean sign and magnitude at least b throughout that radius. Scaling the readout by A makes the calibrated radius tend to zero while the noisy signed output is at least $Ab - r_c(w^{(A)})$. Eventually this exceeds $1/2$, so the clean and noisy clipped scores agree exactly. The readout energy simultaneously grows by A^2. The scale needed may be enormous; this observation gives no polynomial target-mass bound for arbitrary neural representations and cannot justify reporting rescaled networks as evidence about ordinary training.

D Distributional localization with network-dependent exceptions

Stability need not hold on the full cube. The exceptional region can even vary with the network, provided its input-set complexity is controlled. The prior below is an exactly conditioned Gaussian, rather than a mixture of separately normalized Gaussian components.

Let $\mathcal{B} = \{B_1, \dots, B_M\}$ be a family of distinct subsets of $\mathcal{X}$, containing the empty set. The family may be chosen using an independent unlabeled pilot. Draw a further unlabeled validation sample V of size m, independent of the labeled sample S. Fix $0 \le \rho < 1$. Condition the raw Gaussian on the existence of an index j such that

$$\widehat{D}_V(B_j) \le \rho, \qquad |T(z_w(x;\eta)) - T(z_w(x))| \le 1/16 \quad (x \notin B_j,\ \|\eta\|_* \le r_c(w)). \tag{15}$$

Then apply the unchanged training likelihood $f(X_i)z_w(X_i) > 1/2$ at every training example, including examples in the exceptional region.

Theorem D.1 (Localized stability under exact union conditioning)**.** *For targets covered by Theorem 3.1, the posterior with support (15) has Gibbs test-margin error at most $\rho + \varepsilon$, with probability at least $1 - \delta$, whenever*

$$m \geq O\left(\frac{\log M + \log(1/\delta)}{\varepsilon^2}\right), \qquad n \geq \mathrm{poly}(d, L, C, c + c^{-1}, \log M, \varepsilon^{-1}, \log(1/\delta)).$$

One sample event gives the guarantee simultaneously for all $N \geq C$ and $0 < t \leq 1$. No modification of the training likelihood is required.

Proof. Write $a = N/t^2 \geq 1$. By a one-sided Hoeffding bound and a union bound, with probability at least $1 - \delta/2$ over V,

$$D(B_j) \leq \widehat{D}_V(B_j) + u \qquad (1 \leq j \leq M),$$

provided $m \geq [\log M + \log(2/\delta)]/(2u^2)$. The global target cylinder is admissible with the empty exceptional set, so for every V, S the full posterior Q satisfies $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \leq aD_*$.

Assign to each network the first index witnessing (15). Write q_j for its posterior probability and Q_j for the conditional network law. The assignment is measurable because the family is finite. The relative-entropy chain rule gives

$$\sum_j q_j D_{\mathrm{KL}}(Q_j\|\mu_{N,t}) = D_{\mathrm{KL}}(Q\|\mu_{N,t}) + H(q) \leq aD_* + \log M. \tag{16}$$

Thus indices for which $D_{\mathrm{KL}}(Q_j\|\mu_{N,t}) > aK_1$ have total posterior mass at most β, where

$$K_1 = \frac{D_* + \log M}{\beta}.$$

This step avoids assuming that each assigned component separately contains the target cylinder; that assumption would be false in general.

For each retained component, the Gaussian entropy moment bound implies $\mathbb{E}_{Q_j} E_v \leq 4K_1 + 2\log 2$. Set $B = (4K_1 + 2\log 2)/\beta$ and discard its readout tail $E_v > B$, of mass at most β. Denote the resulting conditional law by Q'_j. For any conditioning event E of probability p, KL decomposition gives

$$pD_{\mathrm{KL}}(Q_j(\cdot \mid E)\|\mu_{N,t}) \leq D_{\mathrm{KL}}(Q_j\|\mu_{N,t}) + \log 2.$$

Consequently, for $\beta \leq 1/2$, $D_{\mathrm{KL}}(Q'_j\|\mu_{N,t}) \leq 2a(K_1 + 1)$. All retained components are stable outside their own B_j at the common radius $r = \min\{1/6, c/\sqrt{1 + HB}\}$.

Compress each Q'_j separately, using the common finite class from Lemma 4.1 with $K = 2(K_1 + 1)$. Apply the finite-class relative-Chernoff bound simultaneously to the $M|\mathcal{H}|$ masked ramp losses

$$\ell_{j,h}(x) = \mathbf{1}\{x \notin B_j\}\, \psi(f(x)T(h(x))).$$

This event has probability at least $1 - \delta/2$ over the labeled sample. Outside B_j, training margin consistency and the compression coupling make the candidate's empirical masked loss at most α after averaging. The population comparison adds another α. Therefore every retained component has masked test-margin risk at most

$$3\alpha + \frac{8[\log M + \log|\mathcal{H}| + \log(2/\delta)]}{n}.$$

On the validation event, its unmasked risk costs at most another $D(B_j) \leq \rho + u$. Averaging the components and restoring both discarded masses gives

$$R(Q) \leq \rho + u + 2\beta + 3\alpha + \frac{8[\log M + \log|\mathcal{H}| + \log(2/\delta)]}{n}.$$

Take $u = \varepsilon/4$, $\beta = \varepsilon/8$, $\alpha = \varepsilon/12$, and make the last term at most $\varepsilon/4$. All entropy, radius, and cardinality parameters are polynomial in the stated quantities and independent of N, t. Conditional on the pilot, the same finite masked-loss event works for all widths and scales. Averaging over the pilot completes the proof. □

If the family has VC dimension at most $h \geq 1$, its restriction to the finite Boolean cube has at most $(2^d + 1)^h$ distinct subsets. The theorem therefore applies with $\log M \leq h(d+1)\log 2$, even for a continuously parameterized family. The family must be fixed before the validation and labeled samples, or obtained from an independent pilot. No assertion is made that arbitrary network instability sets have width-independent VC dimension.

A concrete example is a union of neighborhoods of M_0 pilot centers, with separate radii chosen by each network. Its uncovered region is an upper orthant in the fixed vector of distances to the centers and has VC dimension at most M_0. Thus network-dependent local coverage can be handled with polynomial sample complexity. The prior still requires actual stability throughout the covered regions. Pointwise testing of the centers or observed validation inputs alone does not verify that condition.

Corollary D.2 (Using the training inputs to choose the support)**.** *If the full exception family is fixed independently of S, the input locations of S itself may replace the independent validation sample in (15). The resulting input-dependent Gaussian prior, followed by the ordinary training-margin likelihood, has the same $\rho + \varepsilon$ guarantee when n also satisfies the coverage sample bound. No additional unlabeled sample is needed.*

Proof. The coverage bound is uniform over the fixed family, and the masked-loss bound is uniform over its product with the common finite candidate class. Both events hold for the same iid input sample; their failure probabilities can be added without independence. The component entropy and compression arguments are deterministic statements for each realized sample. □

This corollary does not allow an arbitrarily complex exception family to be learned from the same sample without accounting for its complexity. For example, take as exception sets the complements of all unions of at most k Euclidean balls with centers in the Boolean cube. This family is fixed in advance. On that cube each center has at most $d+2$ distinct radius/inclusion choices, including an unused ball, so the family has at most $[2^d(d+2)]^k$ descriptions. Posterior-chosen centers and radii are therefore allowed at a cost $\log M \leq k[d\log 2 + \log(d+2)]$. This formalizes a condition that stability holds on a few input neighborhoods covering most observed inputs, with a statistical guarantee for their population coverage. It does not assert that a small number of such neighborhoods suffices for any particular high-dimensional data distribution.

E A weaker condition based on the gap between training and test margins

The primary condition uses the score clipped at the training margins. The argument admits a further weakening which ignores changes entirely inside the region already counted as a test-margin error. We present this as a mathematical variant; it does not retroactively change the primary condition used in the empirical protocol.

Theorem E.1 (Monotone score transforms)**.** *Fix, independently of the training sample, an odd, nondecreasing and 1-Lipschitz map $F : \mathbb{R} \to \mathbb{R}$ that is constant on $[1/2, \infty)$. Put*

$$A = F(1/2), \qquad B = F(1/4), \qquad \Delta = A - B > 0.$$

Replace the main support condition by

$$|F(z_w(x;\eta)) - F(z_w(x))| \le \Delta/4 \quad \textit{for every } x \textit{ and every } \|\eta\|_* \le r_c(w). \tag{17}$$

Keep the same Gaussian prior, observed training margin $yz > 1/2$, and test error event $yz \le 1/4$. Then the main all-width, all-$\gamma \ge 1$ polynomial sample guarantee remains valid, with additional polynomial dependence on Δ^{-1}. The same replacement is valid in the controlled-exception localized prior theorem, with its $\rho + \varepsilon$ conclusion.

Proof. The existing target cylinder has clean and perturbed signed output larger than one, so all these outputs have transformed score fA. Its support membership and raw Gaussian probability are unchanged. The readout-tail and entropy estimates are also unchanged. After the readout truncation $E_v \le B_v$, use common compression radius

$$r = \min\{1, 2\Delta/3, c/\sqrt{1 + HB_v}\}.$$

On a successful coupling, 1-Lipschitzness of F gives transformed score discrepancy at most

$$\Delta/4 + r/8 \le \Delta/3.$$

At a training input, $yF(z_w) = A$, so the compressed candidate has signed transformed score at least $A - \Delta/3$. At a test-margin error, monotonicity and oddness give $yF(z_w) = F(yz_w) \le B$, so its compressed signed score is at most $B + \Delta/3$. These endpoints are separated by $\Delta/3$. Use the $[0, 1]$ ramp equal to one below $B + \Delta/3$, zero above $A - \Delta/3$, and linear between them. Its slope magnitude is $3/\Delta$. The same finite-class bounded-loss concentration and coupling argument therefore gives the same risk inequality. All common class parameters are polynomial in the old parameters and Δ^{-1}. Masking the ramp outside a controlled exceptional set also gives exactly the earlier localized proof. □

The margin-band score. The particularly natural choice is

$$U(z) = \operatorname{sign}(z) \min\{1/4, (|z| - 1/4)_+\}. \tag{18}$$

It is zero on $[-1/4, 1/4]$, grows linearly toward the training margins, and saturates at $\pm 1/4$ once z reaches $\pm 1/2$. Here $A = 1/4$, $B = 0$ and $\Delta = 1/4$. The support tolerance is still $1/16$, and the common compression radius still satisfies $r \le 1/6$. The separating ramp thresholds are $1/12$ and $1/6$. Thus all constants in the main statistical argument remain unchanged.

The score measures signed excess beyond the test-error threshold. For example, changing an output from 0.1 to -0.1 leaves both transformed scores zero. Both outputs already fail the $1/4$ margin criterion for either label; forbidding this fluctuation is unnecessary for the proof. In contrast, movement from an unconfident output into the training-confidence region remains controlled. Training-margin satisfaction and posterior probability of correct test answers retain their original meanings; the transform only defines the prior's stability condition.

This is genuinely weaker than clipping at $\pm 1/2$: with $T(z) = \operatorname{clip}(z, -1/2, 1/2)$,

$$U(z) = \operatorname{sign}(T(z))(|T(z)| - 1/4)_+,$$

and the map on the right is 1-Lipschitz in $T(z)$. Every network satisfying the primary condition therefore satisfies the margin-band condition at the same radius and tolerance.

There is a precise minimality statement within this class of score transforms. Suppose F satisfies the theorem and retains the full available gap $F(1/2) - F(1/4) = 1/4$. Monotonicity and 1-Lipschitzness imply, with $B = F(1/4) \geq 0$,

$$F(z) = B + z - 1/4 \quad (1/4 \leq z \leq 1/2), \qquad 0 \leq F(z) \leq B \quad (0 \leq z \leq 1/4).$$

Using saturation and oddness elsewhere gives

$$U(z) = \operatorname{sign}(F(z))(|F(z)| - B)_+.$$

Thus stability of any such F implies stability of U at the same radius and tolerance. In this explicitly specified class, the margin-band score imposes the weakest stability requirement while retaining the full training-to-test gap. This is not a claim of optimality among all possible prior conditions.

A raw-output interval certificate $z_w(x; \eta) \in [z - B_x, z + B_x]$ certifies this variant whenever

$$\max\{U(z + B_x) - U(z), U(z) - U(z - B_x)\} \leq 1/16.$$

Existing interval-certified attacks can likewise be evaluated with U without a new attack search. A failure or success of the primary clipped-score condition need not have the same outcome for this variant.

F Why population stability alone does not complete the argument

Small population instability for each network does not itself control instability at training inputs selected by posterior conditioning. This is a statistical obstruction, even when compression is trivial away from the exceptional inputs. The example below concerns an abstract prior; it is not a lower bound for the particular Gaussian network prior.

Proposition F.1 (Sparse exceptional sets can absorb the training sample)**.** *Let $\mathcal{X}$ have M points with the uniform distribution, let $f \equiv +1$, and fix $1 \leq r \leq M/2$. Write $\rho = r/M$. For every subset $B \subset \mathcal{X}$ of size r, let h_B be $+1$ on B and -1 elsewhere, and assign it exceptional set B. The target has empty exceptional set. Put prior mass $q \in (0, 1)$ on f, and distribute mass $1 - q$ uniformly over the h_B. Then every hypothesis has population exception probability at most ρ and has an exact compression outside its exceptions into the two-element class $\{+1, -1\}$.*

For every labeled sample of size $n \leq r/2$, if

$$q \leq \tfrac{1}{2}(\rho/2)^n,$$

the ordinary consistency posterior has Gibbs error at least $(1 - \rho)/2$. The assertion holds for every sample, including samples with repeated inputs.

Proof. Let $s \leq n$ be the number of distinct observed inputs. A bad hypothesis h_B is consistent exactly when all these inputs belong to B. Under the uniform prior on size-r subsets, this has probability

$$p_s = \frac{\binom{M-s}{r-s}}{\binom{M}{r}} = \prod_{j=0}^{s-1} \frac{r - j}{M - j} \geq (\rho/2)^s \geq (\rho/2)^n,$$

because $s \leq r/2$. Therefore the posterior mass of bad hypotheses is

$$\frac{(1 - q)p_s}{q + (1 - q)p_s} \geq \frac{1}{2}.$$

Indeed $q \le p_s/2$ and $q \le 1/2$ imply $(1-q)p_s \ge q$. Every bad hypothesis has true error $1-\rho$. It also has empirical exception probability exactly one, although its population exception probability is only ρ. The two-element compressor simply maps h_B to the constant negative function and leaves the target positive. □

If $q = e^{-aD}$, where $a = N/t^2$ is the width-and-scale factor in the Gaussian target-mass bound, the displayed condition follows whenever

$$aD \ge n \log(2/\rho) + \log 2.$$

Thus a target-mass estimate of this form, a width-independent finite compression class, and per-hypothesis population instability at most ρ do not by themselves imply a width-independent sample bound. An additional structural property of the actual Gaussian prior could still do so; the proposition neither supplies nor rules out such a property.

This obstruction is relevant to interpreting certificates. An independent held-out set can estimate the population instability of one fixed trained network, or of one frozen posterior. That estimate does not bound the training instability of a posterior obtained by selecting among many possible exceptional sets. The controlled-family localized theorem removes exactly this ambiguity by paying for the family of exceptional sets. A trusted input region fixed before labeled training is the simplest special case.

One can also require stability at every observed training input in an input-dependent prior while retaining the ordinary label likelihood. That gives the same mathematical benefit as a robust training constraint: the empirical exception probability is then zero. It changes the posterior conditioning event, however, and should not be presented as a consequence of population stability alone under the original event.

G Analog targets after polynomial architecture overhead

The main all-width theorem uses Boolean circuit width at the specified layered depth. It does not identify that width with the minimum width of an arbitrary real-valued tanh circuit. Nevertheless, the Boolean target assumption permits a useful extension when polynomial changes to the architecture are allowed.

Proposition G.1 (Digital simulation with controlled precision)**.** *Suppose a width-C, depth-L tanh network, with the architecture used in the main theorem, has physical weights bounded by $R \ge 1$ and satisfies*

$$f(x)z(x) \ge 1/2 \qquad (x \in \{-1,1\}^d).$$

Then f has a Boolean circuit using two-input NAND gates, copies and negations, with both size and layered depth bounded by a universal polynomial in

$$d, C, L, \log(2+R).$$

All constant wires and their propagation are included in this bound. In particular, under the bounded-weight convention $R \le C$, this is polynomial in the original input dimension, width and depth.

Proof. Let $H = L-1$, $A = CR+1$, and $T = d+2+L(C+1)$. Choose an integer

$$b \ge \log_2(8T) + L \log_2 A, \qquad \delta = 2^{-b}.$$

Round each weight to the nearest multiple of δ. Compute every affine sum exactly as a rational number, and approximate each resulting tanh value to absolute error at most δ by a number in $[-1,1] \cap \delta\mathbb{Z}$. We justify a polynomial Boolean cost for this operation below.

Let e_s be the maximum coordinate error at hidden layer s. Using the original weights for the propagated error, the rounded activations bounded by one, and 1-Lipschitzness of tanh gives

$$e_1 \le (d+2)\delta, \qquad e_s \le CRe_{s-1} + (C+1)\delta.$$

The exact rational readout then differs from the original real readout by at most $CRe_H + C\delta \le \delta TA^L \le 1/8$. It has the correct sign everywhere, with signed value at least $3/8$. Thus deciding the sign of this rational computation computes f exactly. The needed weight precision is

$$b = O\big(\log(d + LC + 2) + L\log(CR+1)\big).$$

The total binary length of the rounded weights is polynomial in the parameters stated in the proposition.

For completeness, tanh at this precision has a direct polynomial Boolean evaluation algorithm. Its input q is a rational affine sum of numbers with polynomial binary length. By oddness it suffices to handle $q \ge 0$. If $q \ge b+10$, output one; the error is at most $2e^{-2(b+10)} < 2^{-b-4}$. Otherwise set $x = 2q \le 2(b+10)$ and $m = \lceil 16(b+10)^2 \rceil$. Evaluate the rational Taylor sum

$$E_m = \sum_{j=0}^{m} \frac{x^j}{j!}$$

exactly. The tail is at most twice its first omitted term, since the ratio of successive omitted terms is less than $1/2$. The bound $j! \ge (j/e)^j$ then gives

$$0 \le e^x - E_m \le 2\left(\frac{ex}{m+1}\right)^{m+1} \le 2^{-m} < 2^{-b-4}.$$

All numerators and denominators involved have polynomial binary length: a common denominator may be taken as the denominator of x to power m, multiplied by $m!$. Since $E_m \ge 1$ and the derivative of $(u-1)/(u+1)$ is at most $1/2$ for $u \ge 1$, the rational value $(E_m - 1)/(E_m + 1)$ approximates $\tanh q$ to error less than 2^{-b-5}. Rounding it to the nearest multiple of 2^{-b} gives error less than δ and preserves the interval $[-1,1]$. Restore the sign for negative q.

Binary addition, comparison, multiplication and long division on integers of polynomial binary length have Boolean circuits of polynomial size, by the usual digit-by-digit algorithms. This also follows directly by unrolling those finite algorithms into two-input NAND gates. The number of neurons, affine operations and bits in the preceding computation is polynomial. Hence its full Boolean circuit has polynomial size. Its depth is at most its size; making it layered, forwarding still-needed bits by copy gates, and including all rounded-weight constant bits increases size and width by at most another polynomial factor. No real-number oracle or exact evaluation of tanh is used by the Boolean circuit. □

Combining this proposition with the main theorem gives polynomial sample bounds for analog Boolean targets with bounded weights, after a polynomial increase of logical depth and width. The resulting learner must have enough layers for this digital simulation and width at least its layered Boolean width. This is a representational extension of the robust theorem, not a proof at the original analog depth or for every width above the original analog minimum complexity $C_L(f)$. An increase in the learner architecture cannot be silently substituted for those stronger quantifiers.

H An optional weakening: transport of two margin levels

The statistical argument needs separation of the training and test margin levels, rather than stability of every intermediate score. The following optional prior condition expresses precisely that separation. It is a mathematical extension, not the condition used in the primary empirical protocol.

For each network use the same calibrated radius $r_c(w) = c/\sqrt{1 + HE_v(w)}$ as before. Require, for every input x, every perturbation with $\|\eta\|_* \le r_c(w)$, and both $y \in \{-1, 1\}$, the implications

$$yz_w(x) \ge \tfrac{1}{2} \quad \Longrightarrow \quad yz_w(x;\eta) \ge \tfrac{7}{16}, \tag{19}$$

$$yz_w(x) \le \tfrac{1}{4} \quad \Longrightarrow \quad yz_w(x;\eta) \le \tfrac{5}{16}. \tag{20}$$

Because both choices of y are required, this is a label-free condition on the network. It asks a perturbation to preserve a gap between a confident training output and an output counted as a test error. The condition is Borel measurable: extrema over a compact perturbation ball are continuous functions of the weights, and the Boolean input set and label set are finite.

If the original global clean-margin condition $|z_w(x)| > 1/2$ is also retained, the second implication is redundant. The first one for the network's predicted sign guarantees perturbed margin at least $7/16$; negating this inequality gives the second implication for the opposite label, the only label for which it is triggered. In this original globally separated model, the restriction is an ordinary all-layer adversarial margin requirement. The second implication is needed when the theorem permits globally uncertain outputs.

Proposition H.1 (The same bound under two-margin transport)**.** *Replace the clipped-score support restriction in the main theorem by (19)–(20). All its sample bounds and all-width conclusions remain valid, with the same training likelihood, test margin, and posterior-confidence consequences. The same replacement is valid for the localized versions, where the implications are imposed outside the permitted exception set.*

Proof. Use the same common compression radius $r \le 1/6$. On a successful compression coupling, the candidate's raw output differs from a perturbed original output by at most $r/8 \le 1/48$. At an observed training input, $yz_w(x) > 1/2$, so (19) implies

$$yh(x) \ge \frac{7}{16} - \frac{1}{48} = \frac{5}{12}.$$

At a test input where $yz_w(x) \le 1/4$, (20) implies

$$yh(x) \le \frac{5}{16} + \frac{1}{48} = \frac{1}{3}.$$

Thus the same ramp, zero at and above $5/12$ and one at and below $1/3$, has zero training loss on successful couplings and dominates the original test-error indicator on successful couplings. The compression entropy and readout-tail bounds are unchanged.

Every weight in the target cylinder has the correct signed output greater than one both before and after every admissible perturbation. It therefore satisfies both implications for both labels. The original raw Gaussian target-mass lower bound, and hence the posterior entropy bound, remain valid. The rest of the proof is identical. Masking by the controlled exception sets gives the localized version. □

The margin-band score $U(z) = \operatorname{sign}(z) \min\{1/4, (|z| - 1/4)_+\}$ gives a sufficient condition for these implications: if $|U(z_w(x;\eta)) - U(z_w(x))| \le 1/16$, then a clean signed score at least $1/2$ has

perturbed U-score at least 3/16, hence perturbed signed score at least 7/16. A clean signed score at most 1/4 has U-score at most zero, so its perturbed signed score is at most 5/16. Oddness of U handles either label.

The implications allow additional changes at intermediate scores. For example, the pair of scores 3/8 and 11/20 respects both implications for both labels, although its U-score change is 1/8. Such an increase in confidence need not affect either of the two loss-transfer steps. This explains the weakening; it does not establish that trained networks satisfy it globally.

More generally, with training margin m, test margin $u < m$, and a noise tolerance $a < (m-u)/2$, replace the four numerical levels by $m, m-a, u, u+a$. If the compressor's output-rounding error is e and $m - u - 2a - 2e > 0$, the same proof uses a ramp across this positive gap. Its sample dependence is polynomial in the inverse gap. As before, an empirical assessment of this optional condition would require a separate stated protocol; the primary experiments are not relabeled.

I Pointwise support checks with an explicit unlabeled-data cost

There is another way to avoid a global support restriction or a low-complexity family of exceptional regions: use enough unlabeled inputs to pay for unrestricted variation of the instability sets. The price is an unlabeled sample size that depends on width and prior scale. The labeled sample bound remains width independent.

Fix one Gaussian learner, with width N, $0 < t \le 1$, and raw prior $\mu = \mu_{N,t}$. Put $a = N/t^2$. Let $\mathsf{Stable}(w, x)$ denote the primary clipped-score condition at the single input x, with its calibrated radius $r_c(w)$. The margin-band or two-margin conditions may be used instead. Define

$$b(w) = D\{x : \mathsf{Stable}(w, x) \text{ fails}\}.$$

Draw an independent unlabeled sample $U = (U_1, \dots, U_m)$ and a labeled sample $S = ((X_i, f(X_i)))_{i=1}^n$. Condition the raw Gaussian on stability at every input in U and S, then apply the usual likelihood $f(X_i)z_w(X_i) > 1/2$. Thus the support restriction depends on input locations but not on their labels. Optional original global clean-margin conditioning may also be retained. Write $Q_{U,S}$ for the resulting posterior.

Lemma I.1 (Uniform coverage from an unlabeled stability pool)**.** *With probability at least $1 - \delta_U$ over U, every probability law Q supported on stability at all U_j satisfies*

$$\mathbb{E}_Q b(w) \le \frac{2}{m}\left[D_{\mathrm{KL}}(Q\|\mu) + \log(1/\delta_U)\right].$$

The assertion includes laws selected after observing the labeled sample and its labels.

Proof. For a fixed w, the probability of stability at all U_j is $(1 - b(w))^m$. Consequently

$$\mathbb{E}_U \int e^{mb(w)/2} \mathbf{1}\{\mathsf{Stable}(w, U_j) \text{ for every } j\}\, d\mu(w) \le \int e^{mb(w)/2}(1 - b(w))^m\, d\mu(w) \le 1.$$

Markov's inequality bounds the displayed integral by $1/\delta_U$ on an event of probability at least $1 - \delta_U$. On this event apply the entropy variational inequality to the measure with density $e^{mb/2}$ on the empirical stability set and zero outside it. Since Q is supported on that set, it gives

$$\frac{m}{2}\mathbb{E}_Q b \le D_{\mathrm{KL}}(Q\|\mu) + \log(1/\delta_U).$$

Equivalently one may first truncate the logarithmic density below and then pass to the limit. This proves the simultaneous assertion. □

Theorem I.2 (Width-independent labeled complexity with an unlabeled pool)**.** *Under the Boolean-target assumptions of the main theorem, let D_* be its polynomial target-cylinder cost, so that the raw target cylinder has mass at least e^{-aD_*}. For every fixed $N \geq C$ and $0 < t \leq 1$, there is a labeled sample bound polynomial in the same parameters as the main theorem, independent of N and t, such that the following holds. If*

$$m \geq \frac{8}{\varepsilon}\left[aD_* + \log(2/\delta)\right],$$

then, with probability at least $1-\delta$ over U, S, the posterior $Q_{U,S}$ has Gibbs test-margin error at most ε. The usual posterior-confidence consequences follow.

Proof. Every member of the target cylinder is stable on the entire cube and has the correct signed margin greater than one. Thus it belongs to every conditioning event here, for every realization of U, S. The posterior is a raw Gaussian conditioned on an event of probability at least e^{-aD_*}, and therefore

$$D_{\mathrm{KL}}(Q_{U,S}\|\mu) \leq aD_*.$$

Lemma I.1, with $\delta_U = \delta/2$, gives $\mathbb{E}_{Q_{U,S}} b(w) \leq \varepsilon/4$ under the displayed unlabeled sample bound.

The same entropy bound gives the main theorem's readout truncation and width-independent compressor. Choose the discarded posterior mass at most $\beta = \varepsilon/4$ and the coupling failure probability $\alpha = \varepsilon/12$. All posterior draws are stable at the labeled input locations, so the finite candidate law has empirical ramp loss at most α. The same uniform finite-class event over S bounds its population ramp loss by $2\alpha + 8[\log|\mathcal{H}| + \log(2/\delta)]/n$.

At a fresh test input, loss transfer can fail only on a compression failure or when the original network is unstable at that input. Restoring the truncated mass and weighting the conditional instability term by its retained mass gives

$$R(Q_{U,S}) \leq \beta + 3\alpha + \frac{8[\log|\mathcal{H}| + \log(2/\delta)]}{n} + \mathbb{E}_{Q_{U,S}} b(w).$$

Choose n so that the finite-class term is at most $\varepsilon/4$. The four terms then sum to ε. The compression class has the same polynomial logarithmic size as before, independently of N, t, U, S. The two sample events have total failure probability at most δ. □

This is an input-adapted prior with the usual label likelihood. In particular, observing each labeled input also imposes an additional label-free stability constraint; this is a specified change to the learning rule, not a theorem about the original fixed prior. It requires stability only at finitely many specified inputs, so it does not infer region-wide stability from pointwise certificates. It also does not provide a width-independent bound on the total data used: at $t = N^{(1-\gamma)/2}$ its unlabeled cost contains $a = N^{\gamma}$. For each fixed learner the probability statement above is complete. To cover a prescribed countable collection of widths and scales simultaneously, use the one common finite-class event for the labeled sample with failure probability $\delta/2$, and allocate unlabeled failure probabilities δ_j with $\sum_j \delta_j \leq \delta/2$. Use unlabeled sizes $m_j \geq 8[a_j D_* + \log(1/\delta_j)]/\varepsilon$; the labeled sample size is unchanged, and the unlabeled pools may be nested prefixes of one iid sequence. No simultaneous statement over a continuum of changing Gaussian priors is asserted without further argument.

An equivalent semi-supervised protocol chooses the prior before any labels are revealed. Draw one iid pool of $M \geq \max\{m, n\}$ input locations, impose stability on the whole pool, and then reveal the labels of its first n points. The coverage argument uses all M inputs, and the finite-class generalization event uses the first n. These two events need not be independent: each has its stated marginal probability, and their failures are combined by a union bound. The same guarantee therefore holds, with n labels and M input locations.

J From a point certificate to an input neighborhood

A pointwise certificate can sometimes establish stability on a whole region, as required by the localized-prior theorem. The following elementary transfer also makes its quantitative limitation explicit. Write W_{in}^1 for the input columns of the first-layer matrix, excluding the bias column, and define the network-dependent input semimetric

$$d_w(x, x_0) = \frac{\|W_{\text{in}}^1(x - x_0)\|_2}{\sqrt{N}}.$$

Its balls are ellipsoidal cylinders; directions in the nullspace of W_{in}^1 have zero distance.

Lemma J.1 (Transfer of a confident point certificate)**.** *Fix a network, $r' > r_c(w)$, an input x_0, and $p \in \{-1, 1\}$. Suppose that*

$$p z_w(x_0; \eta) > 1/2 \qquad \text{for every } \|\eta\|_* \le r'.$$

Then every x with $d_w(x, x_0) \le r' - r_c(w)$ satisfies

$$T(z_w(x; \eta)) = T(z_w(x)) = p/2 \qquad (\|\eta\|_* \le r_c(w)).$$

In particular, the clipped-score stability condition holds throughout this ellipsoidal cylinder, with zero score change.

Proof. The clean first-layer states differ by

$$b = \tanh(W^1 \bar{x}) - \tanh(W^1 \bar{x}_0), \qquad \|b\|_2/\sqrt{N} \le d_w(x, x_0).$$

A perturbed computation at x is exactly the computation at x_0 with b added to its first-layer disturbance; every subsequent disturbance is unchanged. The combined disturbance has norm at most $\|\eta\|_* + \|b\|_2/\sqrt{N} \le r'$. Apply the assumed margin bound, first to this computation and then to the clean computation at x. □

This construction can be used with a controlled-complexity exception family. On the Boolean cube, a quadratic inequality is a linear threshold in at most $J = (d+1)(d+2)/2$ monomials. Its VC dimension is at most $J + 1$, and hence it defines at most $(2^d + 1)^{J+1}$ distinct subsets of the cube. Complements of unions of at most k ellipsoidal cylinders therefore form a fixed family with

$$\log M \le k(J + 1)(d + 1) \log 2.$$

This deliberately loose bound allows both centers and matrices to vary with the network, and does not depend on network width. To apply Theorem D.1, one must still establish the full center certificates and validate the population coverage of their union. Testing only the centers does not establish either premise.

There is an absolute upper limit on the center radii in this particular construction, for the primary unrestricted disturbance model. Last-layer and scalar-output disturbances can decrease a signed output by exactly $r'\sqrt{1 + E_v}$. Consequently the premise of Lemma J.1 requires

$$r' \le \frac{|z_w(x_0)| - 1/2}{\sqrt{1 + E_v}}.$$

This upper bound is not asserted for the variant that restricts noisy hidden states to the activation range.

A feasibility diagnostic, not a neighborhood certificate. For model 008, a plain tanh network with 32 hidden layers and width 128 trained by minibatch SGD on MNIST digits 3 versus 8, we used the fixed 128 uniformly sampled pilot images from its 1,984-image test split and measured coverage on the other 1,856 images. The pilot indices were chosen before this diagnostic, without selecting for margins or coverage. At target calibration $c = 0.001$, even granting every center the larger calibration $c' = 0.0625$, whether certified there or not, would give the common cylinder radius

$$\frac{c' - c}{\sqrt{1 + HE_v}} = 0.000658742.$$

None of the 1,856 remaining images was covered. Their nearest-center distances in d_w had minimum 0.31845 and median 0.60165; the minimum was 483 times the granted radius. Replacing every center's radius by the absolute upper bound above, minus $r_c(w)$, yielded only 125 covered images, or 6.74%. These are float64 geometric diagnostics, not outward-rounded proofs or distributional coverage estimates. They show that the current certificates and this fixed 128-center transfer construction do not produce useful coverage in this example. They do not bound stability on more general input regions, or imply that the actual stable neighborhoods of the data are small. The diagnostic is recorded in `pilot-neighborhood-diagnostic.json`.

K Gradient descent can be perfectly stable on the data and fragile elsewhere

A global support condition need not hold for a network obtained by ordinary gradient descent, even when the network is robust on the entire data distribution. The following exact construction separates these requirements without numerical attacks or a continuous-input relaxation. It is a counterexample to an automatic guarantee, not a prevalence claim about commonly trained networks.

Proposition K.1 (A gradient-descent trajectory with off-distribution fragility)**.** *Take $d = 2$, width $N = 1$, and $H \geq 1$ hidden tanh layers. Let the data distribution be uniform on $(1, 1)$ and $(-1, -1)$, with label $f(x) = x_1$. Initialize the first-layer affine map as $(x_1 + x_2)/2$, all internal scalar weights as 2, and the readout as $v = 1$. Train all weights and the first-layer bias by full-batch gradient descent on logistic loss, using any finite positive learning rates.*

At every finite iterate, the classifier is correct on the entire data support with signed output at least $\tanh(1) > 1/2$. For every $0 < c \leq 1/100$, it has zero clipped-score change on this support under every simultaneous disturbance of norm

$$\|\eta\|_* \leq \frac{c}{\sqrt{1 + Hv^2}}.$$

The same assertion holds for the margin-band score.

At the valid Boolean input $x_0 = (1, -1)$, however, its clean output is zero and there is an admissible first-layer postactivation disturbance giving

$$z(x_0; \eta) \geq \min\left\{\frac{1}{2}, \frac{c(3/2)^{H-1}}{\sqrt{H+1}}\right\}. \tag{21}$$

Consequently every iterate fails the global clipped-score condition when the second expression in the minimum exceeds $1/16$. Every iterate fails the global margin-band condition when it exceeds $5/16$. For $c = 10^{-3}$, $H \geq 16$ suffices for the former and $H \geq 20$ for the latter.

Proof. Write the first-layer weights as (a, a) and its bias as zero, the internal scalar weights as w_s for $2 \le s \le H$, and the readout as v. At initialization $a = 1/2$, $w_s = 2$, and $v = 1$. The two examples have opposite inputs and labels, and tanh is odd. Their signed outputs are equal. Their loss derivatives with respect to the first-layer bias cancel, and the derivatives with respect to the two first-layer weights are equal. Moreover the signed output strictly increases with each of $a, w_2, \dots, w_H, v$ when they are positive: all factors in the corresponding chain derivatives are positive. The derivative of logistic loss with respect to the signed output is negative. Thus gradient descent preserves zero bias and equal first-layer weights, while increasing a, w_s, v. In particular, throughout the trajectory,

$$a \ge 1/2, \qquad w_s \ge 2, \qquad v \ge 1.$$

The same conclusion holds for minibatches containing equal numbers of the two examples. It does not assume arbitrary stochastic minibatches preserve the symmetry.

At either data input, let $y \in \{-1, 1\}$ be its label. The clean first signed hidden value is at least $\tanh(1) > 3/4$. Induction gives that every subsequent signed hidden value is at least $\tanh(1)$ because $2\tanh(1) > 1$. Hence the signed readout is greater than $1/2$.

For the finite-noise assertion, each scalar disturbance, including the output disturbance, has magnitude at most c. The first noisy signed hidden state is at least $\tanh(1) - c > 0.7$. If the preceding noisy signed state is greater than 0.7, then the next is at least $\tanh(1.4) - c > \tanh(1) - c > 0.7$. Induction gives a final signed output greater than $0.7v - c \ge 0.69 > 1/2$. Clean and noisy outputs therefore lie beyond the same clipping endpoint. Both the clipped score and margin-band score are exactly unchanged on the data support.

At x_0, the equal first-layer weights cancel and every clean hidden state is zero. Inject only at the first hidden layer, with

$$\eta_1 = \frac{c}{v\sqrt{H+1}}.$$

This is admissible because $(1 + Hv^2)\eta_1^2 \le c^2$ follows from $v \ge 1$. All subsequent noisy hidden states are nonnegative. Concavity of tanh on the positive half-line and $\tanh(1) > 3/4$ imply

$$\tanh(2u) \ge \min\{1/2, (3/2)u\} \qquad (u \ge 0).$$

For $0 \le u \le 1/2$, use the chord bound $\tanh(2u) \ge 2u\tanh(1) \ge (3/2)u$; for larger u, use $\tanh(2u) \ge \tanh(1) > 1/2$. Since $w_s \ge 2$, iterating the displayed bound yields

$$z(x_0; \eta) \ge \min\left\{v/2, \frac{c(3/2)^{H-1}}{\sqrt{H+1}}\right\},$$

which proves (21). The clean transformed scores are zero. A noisy output greater than $1/16$ violates the clipped-score tolerance; an output greater than $1/4 + 1/16 = 5/16$ violates the margin-band tolerance. At $c = 10^{-3}$, the latter expression is greater than 0.106 at $H = 16$ and greater than 0.483 at $H = 20$. It increases with H, proving the stated sufficient depths. □

This example also shows why the trusted-region variant is substantive. Taking the trusted region to be the two-point data support gives an exact whole-region stability certificate for every gradient-descent iterate, whereas a full-cube condition rejects all these same networks at sufficient depth. Extending this observation to realistic data requires an actual region-wide certificate or a controlled family of exceptional regions; held-out pointwise success alone is not such a certificate.

L Output scale and membership in the restricted prior

The restriction is a property of a parameterized predictor. It is not a property of its Boolean sign function alone. This matters when comparing networks trained with different losses or readout conventions.

Proposition L.1 (Every strictly signed finite-domain predictor can be rescaled)**.** *Fix a finite input domain and a tanh network whose clean outputs obey* $a := \min_x |z_w(x)| > 0$*. Write*

$$B_s = \prod_{j=s+1}^{H} \|W^j\|_{\mathrm{op}}, \qquad B = c \max\left\{1, \sqrt{\frac{1}{H}\sum_{s=1}^{H} B_s^2}\right\},$$

where an empty product is one. Keep every hidden weight fixed and replace the readout by λv*. If* $\lambda a > 1/2 + B$*, the resulting network satisfies the global clipped-score condition with* zero *score change, at the radius prescribed for its new readout. It also satisfies the weaker margin-band condition.*

Proof. The 1-Lipschitz property of tanh and layerwise propagation give, for every simultaneous disturbance,

$$|z_{\lambda v}(x;\eta) - \lambda z_v(x)| \le \|\eta\|_* \sqrt{1 + \lambda^2 E_v \sum_{s=1}^{H} B_s^2}.$$

At the calibrated radius this is at most

$$c\sqrt{\frac{1 + \lambda^2 E_v \sum_s B_s^2}{1 + \lambda^2 H E_v}} \le B.$$

Thus clean and noisy outputs remain beyond the same clipping endpoint at every input. Both bounded scores are constant there. □

This observation does not supply a polynomial sample theorem for arbitrary small analog circuits. The sufficient λ can be exponentially large in depth, or large because of a tiny minimum output on the finite domain. Its readout energy becomes $\lambda^2 E_v$, and the Gaussian probability cost is charged accordingly. The main theorem instead constructs a robust target implementation with polynomial probability cost. The empirical protocol retains the actual trained readout and does not rescale a saved model to make the condition pass.

Likewise, a larger allowed inverse-polynomial factor in the noise radius is a meaningful calibration choice only when it is stated. The theorem allows such choices because it tracks polynomial dependence on c^{-1}; it does not turn an exponentially small required noise radius into a polynomial-depth guarantee.

M What empirical evidence can establish

The prior's support condition quantifies over every Boolean input and every allowed simultaneous disturbance. A finite-noise certificate at a held-out image proves stability at that image only. A successful adversarial search provides a counterexample when both its disturbance budget and its score change are verified. Failure to find an attack proves neither stability nor membership in the prior. Floating-point sufficient tests are reported separately from interval or exact-arithmetic checks.

There are two distinct empirical eligibility questions. First, does a trained network satisfy the stability restriction? Second, does it satisfy the likelihood, which requires every signed training

output to exceed 1/2? A stable network that has training mistakes is evidence about the condition's occurrence under training, but is not a draw admissible under the noiseless posterior for that training set. Conversely, successful optimization alone provides no stability guarantee. All declared runs, including optimization failures, are retained.

The original all-input clean-margin convention imposes a further requirement beyond training eligibility. The image counterexamples near zero output already fail that clean-margin requirement at their witness input. They refute automatic global stability of the trained parameterization; they are not examples already inside that stronger base support. In contrast, selected exhaustive Boolean-domain positives verify absolute clean margins above 1/2 everywhere, as well as all signed training margins and internal stability. The separate readout rescaling result explains how a finite-domain parameterization can enter both supports, but charges the resulting Gaussian energy cost.

Finite grayscale images and Boolean inputs. An eight-bit pixel $p \in \{0, \dots, 255\}$ normalized as $x = 2p/255 - 1$ can be written exactly as

$$x = \sum_{b=0}^{7} \frac{2^b}{255} B_b, \qquad B_b \in \{-1, 1\}.$$

Composing the first affine layer with this fixed linear encoding represents the same image network as a Boolean-input tanh network with eight times as many input coordinates. All hidden computations and internal disturbance responses are unchanged at corresponding inputs. This observation connects the stability property to a finite image domain; it does not equate the independent Gaussian prior on pixel weights with the independent Gaussian prior on bit weights, and it does not turn a test-set check into a proof over all possible images.

Why a continuous-box counterexample has a different scope. When the input dimension exceeds the first hidden width, one may solve $W^1 \bar{x} = 0$ and sometimes obtain $x \in [-1, 1]^d$. With no later biases, every clean hidden state and the readout then vanish. This exposes amplification that saturation at ordinary data inputs can hide. Such an input is a useful stress test for a continuous-box version of the condition. It is a counterexample to the Boolean-domain condition only if it is a Boolean input, or if a separate valid discrete input is found and verified. Likewise, an interpolated image need not lie on the exact eight-bit grid.

Interpreting calibration. The factor $\sqrt{1 + HE_v}$ is the worst-case noise gain for hidden layers that are individually nonexpansive in Euclidean norm. Hence every such network satisfies the score condition when $c \le 1/16$. Smaller c allows greater amplification relative to this reference. A positive result only at a small radius should not be described as stability to arbitrary percent-level changes of every activation. Both the normalized total budget and the readout calibration are part of the claim.

Global failures motivate studying fixed trusted input regions, chosen independently of the labeled sample, on which the restriction is imposed. If such a region has input-distribution mass at least $1 - \rho$, the statistical proof adds at most ρ to the test error. Proving stability throughout the region remains necessary; merely observing no failures on a finite set of images does not supply that proof.

N Experiments on image classifiers

We tested the support condition on raw parameterizations produced by minibatch optimization. These experiments do not sample from the Bayesian prior or posterior. In particular, a trained

network's test accuracy does not verify the posterior generalization theorem.

Tasks and optimization. The data are the official MNIST and Fashion-MNIST training and test sets. The two-class tasks use all MNIST digits 3 and 8, and all Fashion-MNIST classes 0 and 6. A harder task uses all 60,000 MNIST training images and 10,000 test images with an even-versus-odd label. The main architecture is a plain tanh network with width 128, 256, or 512, hidden depth 4, 16, or 32, a first-layer bias, and a scalar linear readout. There are no later hidden biases and no residual connections. Training uses logistic loss and actual shuffled minibatches of size 128. Vanilla SGD and Adam are distinguished; a small exploratory ReLU cohort uses momentum SGD and lies outside the tanh theorem.

Every cohort has a timestamped protocol fixed before that cohort's training. Later cohorts were motivated by earlier observations, so the whole research sequence is exploratory. All runs, including failed fits and collapsed classifiers, are retained. We use the final prescribed checkpoint and never rescale its readout. Unless separately declared, training lasts 60 epochs. The two depth-32 MNIST SGD seeds also have separately recorded continuations for 180 further epochs, at a fixed lower learning rate; both original models remain in the data. Four later SGD runs probe depths 64 and 128 at width 128, with fixed orthogonal initialization gains 1 and 1.1. All four also receive separately declared 180-epoch lower-rate continuations; the original fixed-budget runs remain in the record.

Continuous pixels are mapped to $[-1, 1]$ in float32; Boolean pixels use threshold 128. Final metrics and ordinary numerical audits cast those same input values and saved weights to float64. Thus auditing does not change the represented input. These are small fully connected image models, rather than modern convolutional networks or transformers.

Eligibility and calibration. For every model we separately record whether *all* training signed outputs exceed $1/2$. High accuracy alone does not meet this likelihood condition. The primary score is clipped at $\pm 1/2$, the allowed score change is $1/16$, and the radius constants are

$$c \in \{0.001, 0.003, 0.01, 0.03, 0.0625\}.$$

The primary constant is 0.001; the rest are a fixed calibration study. The cap 1 is an additional recorded comparison. Neither radius nor readout scale is adjusted to make an individual model pass.

The final collection contains 70 parameterizations: 60 tanh image models, four ReLU comparisons, and six Boolean-domain models. Every training margin exceeds $1/2$ for 13, three, and six of these, respectively, for a total of 22. These counts include every prescribed failed fit and every separately retained continuation.

Pointwise audit and its limits. For each tanh image model the prescribed audit uses uniformly sampled held-out images, the 16 held-out images with smallest absolute clean output, and independent uniform Boolean-cube probes. The second group is deliberately concentrated near the decision boundary. At width 512, the uniform positive-certificate budget is reduced to the first 32 of the same prescribed 128 points. Figure 1 uses common per-model denominators: 32 uniform images, 16 boundary images, and 32 Boolean probes at every radius. Missing or inconclusive certificates remain unresolved. A companion file applies exactly the same display rules to training-margin-eligible models only.

The positive audit propagates finite reachable-state bounds and saturation-aware tanh remainder bounds. Its ordinary pointwise results use float64 evaluations of sufficient inequalities. Attacks optimize both raw-output directions over simultaneous perturbations of every hidden layer and the

Cohort	Runs	All train margins $> 1/2$	Test accuracy (%)
Initial continuous images	24	3	49.95–99.55
Initial Boolean images	8	1	49.95–99.19
Initial width-256 extension	4	1	49.95–99.29
Width-512 extension	4	3	87.10–99.50
All MNIST, even versus odd	4	4	98.92–99.03
ReLU comparisons	4	3	86.90–99.13
Spectral-projection intervention	4	0	86.20–98.95
Additional matched reference	1	0	87.15
Projected-SGD continuation	1	0	98.83
Depth-32 SGD continuations	2	1	99.14–99.24
Depth-64/128 extension	4	0	98.79–98.99
Depth-64/128 continuations	4	0	98.74–98.94
Enumerated Boolean domains	6	6	92.86–100.00
Total	70	22	

Table 1: Every declared training run is retained. Accuracy ranges are descriptive extrema over the specified cohort, not confidence intervals. Continuations are additional parameterizations of retained original runs. Training-margin eligibility is separate from robustness.

	Test accuracy (%)		Minimum train margin	
Task, depth	SGD	Projected	SGD	Projected
MNIST even/odd, $H = 4$	98.94	98.95	2.69	-3.59
MNIST even/odd, $H = 16$	98.92	98.87	4.49	-1.66
Fashion 0/6, $H = 4$	87.15	87.10	-4.96	-5.44
Fashion 0/6, $H = 16$	86.85	86.20	-3.41	-4.10

Table 2: Matched spectral-projection intervention. The models share initialization and minibatch order, and receive the same gradient-step budget. Raw readouts are not rescaled.

scalar output, with two starts and 100 projected gradient steps per direction. The main attack subset is the first 16 points of each group; points without an attack search are still certified only when a positive bound succeeds. All observed violating tuples are saved and re-evaluated strictly inside the prescribed budget. Selected positive and negative results receive independent exact-rational or outward interval verification, as described separately. An unsuccessful attack is not a positive certificate, and an unsuccessful certificate is not a counterexample. Pooled input/model counts reuse data and are not independent statistical observations.

Among the 13 training-margin-eligible tanh image models, the primary radius receives sufficient float64 certificates on all 1,184 uniformly held-out, 208 boundary held-out, and 832 random Boolean input/model pairs in the prescribed positive-certificate subsets. At $c = 0.01$, the corresponding counts are 1,181, 150, and 820. These finite sampled successes coexist with the separately targeted global counterexamples below. An independent audit compares all 28,800 matched attack/certificate evaluations across the 60 tanh image models; none exceeds its sufficient bound beyond 10^{-8}.

A same-model positive and negative result. The Boolean-input MNIST model `029` has hidden depth 16, width 128, was trained by vanilla SGD, and has minimum training signed output above 4.23 and test accuracy 99.19%. Its raw output has a global all-real-input certificate at $c = 0.001$. At $c = 0.01$, exact-budget checks and outward interval arithmetic verify clipped-score

violations on both a held-out Boolean image and a uniform Boolean probe. The score changes exceed 0.0845 and 0.0855, respectively. In both attacks every disturbed hidden state remains in $[-1, 1]$. Thus the calibration constant matters even for a fitted classifier, and the larger-radius failures are not caused by allowing hidden states outside the activation range.

The full-MNIST even-versus-odd model `104`, with hidden depth 4 and width 256, fits every training margin above 2.68, reaches 98.94% test accuracy, and has an exact-rational global certificate at $c = 0.003$: its raw output changes by at most $0.03230 < 1/16$. This certificate covers every real input, not merely the recorded images. The statement concerns its trained parameterization; it does not identify its training process with Bayesian inference.

The matched-task depth-16, width-256 model `105` illustrates the limitation of a universal support claim. Exact integer/rational verification puts all 60,000 training signed margins above 4.48157. Certified clean-output intervals give 9,892 correct predictions among the 10,000 test labels, confirming 98.92% accuracy. Yet it has a rigorously verified violation already at the primary constant $c = 0.001$. A synthesized 8-bit image with mean ink about 0.172 and 64.2% black pixels has clean output approximately -0.00163 and perturbed output -0.15496, giving clipped-score change above 0.1533. Every disturbed hidden state remains in $[-1, 1]$. This is a constructed off-distribution image, not a held-out natural image, and its encoding into Boolean pixel bits is discussed separately. Thus fitting every training margin does not by itself enforce the proposed global condition.

A later, separately declared search addresses the objection that the first witness starts near the decision boundary. On another synthetic 8-bit input, the same model has clean output in $[0.53934, 0.53935]$, above the training margin. A hidden-only disturbance at $c = 0.001$ lowers it to $[0.38661, 0.38662]$, below $7/16$. The scalar output disturbance is exactly zero, and every disturbed hidden coordinate remains inside $[-1, 1]$. Exact budget and outward interval checks establish this for both rational pixel normalization and the original float32 preprocessing. Thus the global failure also affects a confident positive output and violates the optional one-sided margin-gap restriction; no true label is assigned to this constructed image.

This failure is sharply separated from the observed image distribution. A separately declared follow-up checked *every* official held-out image for selected models. For model `105`, all 10,000 images receive a sufficient pointwise certificate at both $c = 0.001$ and $c = 0.003$. At the latter constant, 9,960 pass by combining the global raw-output bound with clipping, and the remaining 40 pass using local finite-radius bounds. Both original depth-32 MNIST SGD seeds similarly certify all 1,984 test images at both constants, despite their failed training-margin fits. These initial whole-pool sweeps evaluate sufficient inequalities in float64 on a finite held-out set; they do not cover every natural image or the whole input cube. Across all six prescribed follow-up models (`005`, `008`, `009`, `029`, `105`, and `107`), all 27,936 held-out input/model pairs are certified at both constants. Eleven initially unresolved points for model `107` at $c = 0.003$ were subsequently certified by stronger finite-radius enclosures; that refinement history is retained.

Model `105` subsequently receives an exact upgrade at the primary constant $c = 0.001$: exact integer/rational arithmetic and outward interval bounds certify every one of its 10,000 official test inputs, using their original float32-normalized values. A global bound covers 9,988 inputs and local bounds cover the remaining 12, with maximum certified clipped-score change below $0.017653 < 1/16$. This quantifies over every allowed simultaneous disturbance at each of those inputs. It does not certify unknown images or the population distribution. The result at $c = 0.003$ and the other models' whole-pool results retain their float64 qualification. The same exact method also verifies the training-margin statement above and clipped-score stability at all 60,000 training inputs at $c = 0.001$. Thus this model has verified likelihood eligibility and verified stability on both official data sets while still failing the global input-domain condition.

An explicit spectral-norm intervention. Four further vanilla-SGD runs project every hidden-to-hidden matrix to operator norm at most 0.99 after each epoch, including after the final update. The first layer and readout are untouched. Each has a matched unconstrained run with the same initialization, minibatch order, data, number of gradient steps, and learning-rate schedule. The nonexpansive certificate then guarantees score stability up to $c = 1/16$ when the saved internal norms are verified below one. This is an intervention in training, rather than evidence that ordinary SGD spontaneously enforces the condition. Its accuracy and training margin cost must be assessed alongside the guaranteed stability. The matched MNIST test accuracies change from 98.94% to 98.95% at depth 4, and from 98.92% to 98.87% at depth 16. Fashion test accuracy decreases by 0.05 and 0.65 percentage points, respectively. All four projected runs retain training errors and hence fail the hard training-margin likelihood. They demonstrate useful classification under an enforced support restriction, not fitted examples under that likelihood. The separately declared 240-epoch continuation of the depth-16 full-MNIST projected run still has 154 training errors, compared with 169 initially, and test accuracy 98.83%. Thus the fixed longer optimization budget does not resolve the training-likelihood failure. All five projected parameterizations nevertheless have exact-rational global certificates at $c = 1/16$, including the continued model, whose uniform raw-output variation bound is below 0.058087.

A polynomial-depth calibration variant. After observing the fixed-radius failures, we also screened the explicitly exploratory choice $c_H = 0.001/H$. This weakens the radius by one additional power of depth; a theorem polynomial in $c + c^{-1}$ continues to have polynomial depth dependence under this substitution. For the same model `105`, exact-rational operator bounds certify global raw-output variation at most 0.023658 at $c_H = 1/16000$, so it passes this weaker condition on every real input. This does not undo its failure at the original fixed constant. The larger-depth cohort tests this variant further, without inferring an asymptotic law from finitely many trained networks. The fitted depth-32 continuation `301` yields the same distinction: an interval-verified primary-radius violation coexists with an exact global raw-variation bound below 0.017821 at $c_H = 1/32000$. Its minimum training margin is 2.41066 and test accuracy is 99.244%; the other prescribed continuation still has three training errors.

Reduced-precision diagnostic. We additionally round hidden activations to IEEE float16 or bfloat16, leaving weights fixed, on the official test set. Hidden-only rounding and hidden-plus-output rounding are separate modes. The realized joint noise norm is recorded and compared with each calibrated radius. This is one deterministic perturbation map, not a test of every allowed perturbation and not a worst-case certificate. In particular, successful rounding can occur even when its realized noise lies outside the certified ball. The smallest calibrated ball should not be equated with ordinary low-precision arithmetic: the realized rounding budgets can exceed it even when the classifier's clipped score barely changes. The complete diagnostic reports both the score changes and the actual noise-budget coverage.

Exhaustive Boolean audit. Six additional networks use the exact domain $\{-1, 1\}^9$, with target either majority of the nine bits or majority of the three majorities on disjoint groups of three. A fixed permutation selects 400 training and 112 held-out inputs. Width is 64, hidden depths are 4, 16, and 64, and every model receives the same prescribed 500-epoch SGD schedule. These models are reported separately from the image cohort. The audit enumerates all 512 inputs at the original five radius constants and at $c_H = 0.001/H$. Thus successful coverage concerns the entire Boolean target domain, even when a sufficient bound uniform over all real inputs is inconclusive. Domain coverage

Format	Rounded values	Sign flips	Score changes $> 1/16$	$c = .001$	$c = .0625$
float16	Hidden	1	0	0.00%	87.52%
float16	Hidden and output	1	0	0.00%	32.28%
bfloat16	Hidden	19	54	0.00%	1.22%
bfloat16	Hidden and output	19	53	0.00%	0.05%

Table 3: Deterministic activation-rounding diagnostic on 175,520 held-out input/model pairs across 60 tanh image models. The last two columns give the fraction whose realized joint noise lies inside the corresponding calibrated ball. The clipped score has cap 1/2. Failures to fit the training likelihood are retained; no adversarial or distribution-wide claim follows from this fixed rounding map.

is exhaustive, but the arithmetic is still float64 unless separately verified exactly. All six runs fit the training margin empirically. The initial exhaustive audit certifies all vertices at $c = 0.001$ for five models, and 511 of 512 for the remaining depth-64 recursive-majority model; all six pass on every vertex at c_H. A separately recorded stronger bound resolves the last vertex, enclosing its noisy raw output in $[-2.45149, -2.36141]$ and hence giving zero clipped-score change. Thus all six ultimately pass on the entire Boolean domain at the primary constant. These finite examples test the support condition, not an empirical sample-complexity law.

The predeclared first exact verification selected depth-4 majority model `500` at $c = 0.03$. Exact interval arithmetic verifies all 400 training signed margins above 1/2 and all 512 clean absolute outputs above 4.72037. Combined with an exact uniform raw-output variation bound below 0.162903, this proves that every clipped score is constant throughout the allowed noise ball on the entire Boolean domain. The raw variation bound exceeds 1/16, so this certificate uses clipping rather than merely a small raw-output bound; it does not show that the network violates the raw-output stability condition.

A deeper exact positive is model `504`, the depth-64 majority network. All 400 training margins are interval-verified, and the network correctly labels 511 of the 512 Boolean vertices. At the primary constant $c = 0.001$, an exact finite partition and reachable-tube calculation proves zero clipped-score change on every vertex for every allowed simultaneous perturbation. The scalar global raw-output bound is approximately 13.4145 and is insufficient on its own. Thus this is a fitted deep SGD example satisfying the actual finite-domain support condition, certified using the states reached by its inputs. The depth-64 recursive-majority model `505` also receives an exact all-domain certificate at $c = 0.001$, combining 511 reachable-tube cases with an exact covariance calculation for the remaining vertex. Every training margin is interval-verified above 7.47388; it correctly labels 504 of the 512 vertices. Both deep examples thus satisfy the support condition without requiring perfect generalization.

The experiment directory preserves protocols, raw weights, dataset hashes, per-example outputs, failed fits, all saved attack witnesses, certificate records, and plotting code. Exact proofs, pointwise numerical evidence, and empirical training outcomes have distinct qualifications throughout.

O Independent finite-radius verification of trained networks

The aggregate empirical audits above use numerical sufficient inequalities and numerical attacks. This section separates out selected positive and negative examples checked by exact rational arithmetic and outward interval evaluation. A certificate for all real inputs is distinguished from an exhaustive certificate on a specified finite Boolean domain. Both quantify over every allowed disturbance. A

Target	H	Min. margin	Test acc.	$c = .001$	$c = .001/H$	All-input c_*
Majority9	4	4.99	100.00%	512/512	512/512	0.0625
Majority9	16	6.94	100.00%	512/512	512/512	0.0625
Recursive majority	4	4.04	100.00%	512/512	512/512	0.0625
Recursive majority	16	6.56	96.43%	512/512	512/512	0.03
Majority9	64	7.61	99.11%	512/512	512/512	0.001
Recursive majority	64	7.47	92.86%	512/512	512/512	0.001

Table 4: Exact finite Boolean domains, with float64 evaluation of sufficient finite-radius inequalities. Certified counts cover all 512 inputs; c_* is the largest original-grid constant for which every input passes, a lower bound on the valid radius rather than an estimate of its maximum. The minimum margin is over the 400 training inputs. One initially unresolved vertex at $c = .001$ receives a separately recorded stronger certificate. A failed sufficient certificate remains unresolved; it is not a counterexample.

pointwise certificate covers every disturbance at one specified input, while a negative witness is one verified input–disturbance pair. These scopes are kept distinct.

Global certificates for raw trained models. The global restriction is satisfied by some ordinary, unconstrained trained networks. A width-256, depth-4 tanh network trained by minibatch SGD on all 60,000 MNIST images for even-versus-odd classification has test accuracy 98.94%, no training errors, and minimum signed training output exceeding 2.6876. Its raw saved weights satisfy the global condition at $c = .003$: for every real input and every allowed simultaneous disturbance, the raw output changes by at most $0.032296 < 1/16$. This global bound was verified using exact rational operator-norm certificates and a rational final gain comparison, rather than relying on a floating-point spectral estimate.

There are also eligible depth-16, width-128 examples at $c = .001$: the selected SGD models for continuous and thresholded-Boolean MNIST 3-versus-8 inputs have minimum training margins exceeding .692 and 4.236, respectively, test accuracy 99.29% and 99.19%, and rigorously verified global raw-change bounds .060795 and .062230. These use raw trained weights, without output rescaling. Their global certificates apply to every real input, so imply either clipped-score convention. The radius constant matters: for the latter Boolean model we independently verified violations at $c = .01$ on both a held-out image and a Boolean cube probe, with clipped-score changes exceeding .08450 and .08553, respectively. The successful small-radius certificate and larger-radius failure are compatible.

An exploratory inverse-depth calibration. A later, separate calibration sets $c_H = .001/H$. For the depth-16 full-MNIST model `105`, which has a verified global failure at the original $c = .001$, exact rational spectral bounds prove global raw-output variation at most $.02365730 < 1/16$ at $c_H = 1/16000$. Thus this same trained parameterization passes a weaker, explicitly depth-dependent support condition. This is not a revision of the original protocol or evidence that a fixed c works uniformly in depth. Since the theoretical bound has polynomial dependence on $c + c^{-1}$, such a polynomial depth calibration preserves a polynomial formal sample bound. The training-eligible depth-32 continuation `301` also has an exact global raw-change bound .01782087 at $c_H = 1/32000$, despite its verified failure at fixed $c = .001$.

Projected SGD at a larger fixed radius. All four predeclared projected-SGD models (depths 4 and 16, width 256, on full MNIST parity and Fashion-MNIST shirt-versus-T-shirt) have exact rational certificates that every internal matrix has operator norm below one. Their global raw-output change bounds at $c = 1/16$ are respectively .061573, .058087, .061573, and .058087, all below 1/16. The corresponding test accuracies are 98.95%, 98.87%, 87.10%, and 86.20%. None attains every training margin above 1/2: this is evidence that the support condition can coexist with useful trained prediction, not an example satisfying the paper's exact interpolation likelihood. The separately declared longer-training continuation `600` retains an exact global bound .058087 at the same radius, but still has 154 training errors after 240 additional epochs, with 98.83% test accuracy. Thus all five projected parameterizations have verified global stability, and none is presented as satisfying the exact training-likelihood event.

Exhaustive Boolean-domain certificates. A separate, predeclared cohort trains width-64 tanh networks on 400 of the 512 vertices of $\{-1, 1\}^9$, with the remaining 112 vertices held out. For the protocol-selected depth-4 majority network `500`, exact arithmetic and outward intervals certify the clipped-score condition on *every* Boolean input at $c = .03$. All 400 training margins are independently interval-verified above 1/2, and the network correctly predicts all 512 labels. The global raw-change bound is $.162903 > 1/16$, whereas every clean absolute output exceeds 4.72037. Consequently the clipped score is exactly constant under every permitted disturbance at every vertex. A separate auditor replayed the saved matrix factors, all 512 interval trajectories, and all 400 training inequalities.

A subsequent, separately recorded check gives the same exhaustive conclusion for the depth-16 majority network `501` at the largest primary radius $c = 1/16$. Its raw-change bound is 3.474718, while every clean absolute output exceeds 5.12034; all 512 labels and every training margin are correct. These finite-domain certificates use the elementary implication

$$|z(x)| \geq m, \qquad |z^{\eta}(x) - z(x)| \leq B \leq m - \tfrac{1}{2} \quad \Longrightarrow \quad \mathrm{clip}_{1/2}(z^{\eta}(x)) = \mathrm{clip}_{1/2}(z(x)).$$

They therefore certify the actual clipped-score condition even when the stronger raw-output-change condition is not certified. Their scope is the entire finite Boolean domain, not all real inputs.

A depth-64 example requiring reachable-state bounds. The first depth-64 majority network, `504`, also satisfies the condition on *all* 512 Boolean vertices at the primary $c = .001$. Every one of the 400 training margins is interval-verified above 7.61170, and every clean absolute output on the full cube exceeds 5.64786. Thus this network also satisfies the original all-input clean-margin restriction, not only the newer clipped-score restriction. Exact interval label comparisons give 511/512 correct predictions on the full cube, including 111/112 held-out predictions; the remaining vertex is a confidently wrong prediction, not an omitted or ambiguous case. Its global raw-change bound, 13.414533, cannot establish this conclusion, even with output clipping. Instead, 52 finite groups of vertices share rigorously bounded reachable-state tubes. Each group's disturbed outputs remain beyond the corresponding clipping threshold, with the smallest verified surplus exceeding 1.40770. Thus the clipped score is exactly invariant for every allowed disturbance at every vertex. This is an exhaustive finite-domain certificate for an ordinary SGD-trained depth-64 network, rather than a sampled-input audit or a claim about every real input.

For completeness, sharing a tube across finitely many inputs does not alter the underlying proof. At each layer and coordinate, replace the clean preactivation's distance from zero by the minimum of that distance over all vertices in the group. A common prefix-energy bound then encloses every group's noisy trajectory, so the same derivative and matrix-norm bounds apply to each vertex. The

grouping only reduces computation. Every input is still checked, and any unresolved input would prevent a full-domain conclusion. The certificate uses exact integer matrix arithmetic, outward interval tanh, saved norm-factor proposals, and explicit source hashes; it is not a Lean proof. A separate auditor regenerated all 512 clean interval trajectories, checked all 52 groups and 3,216 saved matrix bounds, and replayed every training inequality. The three worst groups were additionally checked with the original pure-integer matrix-norm verifier.

A depth-64 nonlinear target. The corresponding recursive-majority network `505`, trained on majority of three disjoint three-bit majorities, also satisfies the clipped-score condition on the entire 512-vertex cube at $c = .001$. Every one of its 400 training margins is interval-verified above 7.47388, and every clean absolute output exceeds 2.40649, so the original all-input clean-margin condition holds as well. Its full-domain accuracy is 504/512; the held-out accuracy is 104/112. All eight errors remain in the verified domain.

For this network, the shared reachable-tube proof establishes 511 of the 512 cases. The remaining vertex, index 325, is explicitly retained as unresolved in that component certificate. A separate finite-radius proof propagates the exact clean-response covariance and bounds the nonlinear Taylor remainder in state norm. It encloses the noisy output at this same vertex in $[-2.586691, -2.226290]$, proving that its clipped score is also invariant. The composite certificate records the union of these two proofs, with model, radius, source, and component hashes; the input domain never changes. The exceptional-vertex proof was independently replayed, including all 126 exact matrix-norm bounds. A further independent replay regenerated all 512 clean trajectories, all 122 groups and 7,585 local matrix bounds, and every training margin, then checked the composite coverage. The state-covariance inequality used for this refinement is given below.

O.1 The global condition can fail on valid finite-grid inputs

A universal prior-support restriction and a statement about typical data are different claims. The distinction persists after clipping at the training margin, even for a network that fits all training examples with substantial margin. Our strongest example is an ordinary minibatch-SGD tanh network of hidden depth 16 and width 256, trained on all 60,000 MNIST images for even-versus-odd classification. It has 98.92% empirical test accuracy, and a separate exact computation verifies every signed training output above 4.4815762. The finite-radius counterexample below is independently verified. The saved weights are used without rescaling.

The left image in Figure 3 is a valid 28×28 eight-bit grayscale input. At the smallest tested radius constant $c = 10^{-3}$, its clean and disturbed real outputs are enclosed around

$$-0.00162845134 \longmapsto -0.15496440197.$$

The resulting score change exceeds $0.15333 > 1/16$ after clipping at either $1/2$ or 1. The image has a black two-pixel border, mean ink 0.17200, and 64.2% black pixels. It is a synthetic off-data image, not an image to which we assign a natural digit label. Its pixel p_j is interpreted as the exact rational input $x_j = 2p_j/255 - 1$; we also independently checked the same witness under the original float32 pixel preprocessing used for training.

The disturbance budget was checked using exact integer-rational arithmetic, and the two network evaluations used 100-decimal outward interval arithmetic with the saved dyadic weights. In particular,

$$(1 + HE_v)\left(N^{-1}\sum_{s=1}^{H}\|\eta_s\|_2^2 + \xi^2\right) < 10^{-6}.$$

The intervals verify that every perturbed hidden coordinate remains in $[-1, 1]$: their absolute values are all below 0.061110. Thus requiring disturbances to leave activations within their original range does not remove this counterexample. Its scalar output disturbance has absolute value below 7.3×10^{-10}. Removing it decreases the noise budget and changes either clipped score by at most that amount, so hidden disturbances alone still give a violation. These are finite-radius witnesses, rather than conclusions drawn only from a large Jacobian or from an unverified optimization output. They trust the interval-arithmetic implementation and are not Lean formalizations.

The right image provides a second example in a depth-32, width-128 SGD model for MNIST 3 versus 8, with test accuracy 99.09%. Its output changes from -0.0034068096 to -0.3168576499 at the same $c = 10^{-3}$, a clipped-score change greater than 0.31345. All perturbed hidden coordinates have absolute value below 0.128211. This second model has three training errors, so it is evidence about the naturalness of a support restriction rather than an additional example in the exact training-likelihood event. A related near-gray eight-bit witness for this model has score change greater than 0.30799. The low-ink examples exclude an explanation specific to a uniformly gray background, while remaining clearly off-data.

Each exploratory search had a fixed budget declared before that search. For the low-ink examples we solved an ink-minimizing linear program, subject to zero first preactivations and a black border, generated 4,096 stochastic roundings to the pixel grid using a fixed seed, and attacked the 16 candidates with smallest absolute clean output. We then verified selected witnesses independently. This protocol proves existence, not the prevalence of instability under the data distribution. In separate probes preserving empirical one-pixel foreground probabilities and on interpolations of opposite-class test images, some violations were verified at $c = 10^{-2}$; failure to find violations at smaller radii was not treated as a certificate.

A precise Boolean interpretation. The finite-grid issue is not inherently a continuous-input phenomenon. Write each pixel in bits $p_j = \sum_{b=0}^{7} 2^b q_{jb}$, with $s_{jb} = 2q_{jb} - 1 \in \{-1, 1\}$. Since $\sum_b 2^b = 255$,

$$\frac{2p_j}{255} - 1 = \sum_{b=0}^{7} \frac{2^b}{255} s_{jb}.$$

Replacing row j of the first-layer matrix by eight rows $(2^b/255)W_{0,j}$ gives a network on $8 \cdot 784 = 6272$ Boolean inputs with exactly the same outputs, hidden trajectories, readout energy, and internal-noise budgets. Its depth and width are unchanged. Thus every rational-pixel witness is also an exact Boolean-input witness for this linearly decoded network. With the usual dimension-weighted first-layer Frobenius energy, the non-bias weight contribution grows only by

$$8 \sum_{b=0}^{7} (2^b/255)^2 = \frac{8(4^8 - 1)}{3 \cdot 255^2} < 2.69.$$

The first-layer bias is unchanged but its dimension weight grows eightfold. Consequently total first-layer energy, including the bias, increases by at most a factor of eight. This equivalence uses exact rational pixel normalization. The software used to train and evaluate the original model normalizes pixels in float32; the original-preprocessing witnesses were checked separately, and the held-out empirical audits use precisely those float32-normalized inputs. We do not identify the two normalization maps as exact equalities.

An optional margin-band score. As a separate, post-protocol comparison, consider $U(z) = \operatorname{sign}(z) \min\{1/4, (|z| - 1/4)_+\}$, which ignores changes entirely inside the already-failing test-margin

band. The original depth-32 low-ink witness still violates its 1/16 condition, with U-score change greater than 0.0668576; its near-gray witness does not. For the fitted full-MNIST depth-16 model, the zero-centered witness lies entirely within the ignored band. We therefore declared a new, explicitly variant-specific exploratory protocol: move the continuous low-ink input toward clean output 3/8, generate 4,096 pixel-grid roundings, retain the 16 closest to 3/8, and apply the same internal-noise attack budget. The independently verified witness has

$$z = 0.3628586342 \longmapsto 0.5158098496, \qquad |\Delta U| > 0.1371413 > 1/16$$

at $c = .001$. All disturbed hidden coordinates have absolute value below 0.089882. Here too the scalar output disturbance is below 7.3×10^{-10}, so the same 1-Lipschitz argument gives a hidden-disturbance-only violation. This secondary result shows that the weaker score can move the fragile input region without eliminating the global obstruction; it does not change the primary clipped-score protocol.

The predeclared longer-training check also supplies a depth-32 example. Of the two fixed continuations, model `301` fits all 11,982 training examples with minimum signed output above 2.4106 and achieves 99.24% held-out accuracy. A valid low-ink, black-border 8-bit image has interval-verified output change $-.00895437 \mapsto +.30415992$ at $c = .001$, hence clipped-score change exceeding .31311. Every noisy hidden state remains in $[-1, 1]$. The other continuation retains three training errors; both outcomes and both prescribed searches are retained. Training metrics in this paragraph are ordinary float64 evaluations, whereas the displayed counterexample has an exact rational noise-budget check and outward interval forward evaluation.

A still weaker, exploratory implication-only condition asks that, for both labels y, $yz \geq 1/2$ imply $yz^\eta \geq 7/16$, and $yz \leq 1/4$ imply $yz^\eta \leq 5/16$. Negating the already saved perturbation for model `301`, without any new optimization or input selection, yields the interval-verified output $-.32205298$ from the same clean value $-.00895437$. With $y = -1$, the second implication fails. Its margin-band score also changes by more than $.07205 > 1/16$. The exact noise budget is unchanged by negation and every disturbed hidden state remains in $[-1, 1]$. This auxiliary check concerns later proposed variants; it does not change the primary score convention or empirical protocol.

Failure starting from a confident output. A further, separately declared finite-budget check targets clean output .55 in the fitted full-MNIST model `105`. The resulting valid eight-bit, low-ink input has independently replayed interval enclosures around

$$z = 0.539345460246 \longmapsto z^\eta = 0.386611305539 \qquad (c = .001).$$

Here the scalar output disturbance is exactly zero. The exact rational hidden-noise budget passes, and every disturbed hidden coordinate has absolute value below .101215. Thus the first implication above fails for $y = +1$: a clean signed output above $1/2$ falls below $7/16$. Both the clipped score and the optional margin-band score change by more than $.113388 > 1/16$. The same saved disturbance was independently verified under the original float32 pixel preprocessing. The input is synthetic and has no assigned digit label; the positive clean output establishes the implication's premise. This check refutes an explanation of the observed instability based solely on clean outputs near zero. It does not show that this image model satisfies the original all-input clean-margin restriction, which its other near-zero inputs already violate.

O.2 Exact stability on the complete held-out image set

For the same fitted full-MNIST depth-16, width-256 model `105`, we can verify the primary clipped-score condition on *every one of the* 10,000 *official held-out images* at $c = .001$. This includes

incorrectly classified images; the certificate concerns stability, not correctness. It uses the exact float32-normalized inputs of the training and evaluation pipeline and quantifies over every allowed simultaneous hidden/output disturbance at each such input.

A separate exact pass also verifies *all* 60,000 *signed training margins*, with a minimum certified lower bound exceeding $4.4815762 > 1/2$. Every training input satisfies the clipped-score condition at $c = .001$ as well. Thus, for this particular image model, training-likelihood eligibility is verified by exact arithmetic rather than inferred only from the float64 training metrics. Exact sign comparisons against every archived test label also confirm 9,892 correct and 108 incorrect predictions, with no ambiguous cases, reproducing the 98.92% test accuracy.

A guarded integer computation with rigorously enclosed activation-table and quantization errors gives a clean-output interval at every image, with uniform scalar error below .007149522. Combining these intervals with an exact global disturbance bound certifies 9,988 images. The remaining 12 inputs are explicitly retained as unresolved in that component record. Separate 160-bit interval clean trajectories, exact reachable-tube matrix bounds, and signed Taylor-remainder certificates prove those same 12 cases. The largest certified clipped-score change over the complete set is below $.01765217 < 1/16$. A hash-checked composite record verifies that the two components cover precisely indices $0, \ldots, 9{,}999$, with the same model, input archive, and radius.

The integer clean-output computations were independently repeated, reproducing all 10,000 test and 60,000 training rational row records exactly. A separate replay also reproduced every exact rational row of all 12 local certificates, and the complete-set composition was independently checked. The worst local case received an additional replay of every matrix bound using the original pure-Python-integer verifier. These are exact integer/rational and outward-interval sufficient proofs, not ordinary float64 sufficient tests and not Lean formalizations. The analogous larger-radius $c = .003$ audit remains numerical.

This finite-set result coexists with the independently verified synthetic-input failures at the same $c = .001$, including the confident-input witness above. It therefore establishes a precise distinction between stability on every image in the official test set and stability over every possible image. It does not prove uniform stability on a neighborhood of those images or a distributional coverage guarantee.

O.3 A limited test of a smooth-image trusted region

The preceding low-ink images are visibly atypical. We therefore tested a separate, simple image restriction based on anisotropic total variation,

$$\mathrm{TV}(p) = \sum_{(i,j)\in\mathcal{E}} |p_i - p_j|, \qquad p \in [0,1]^{784},$$

where $\mathcal{E}$ consists of horizontal and vertical nearest-neighbor edges. The 10,000 official MNIST test images have median TV 102.55, 99th percentile 143.74, and maximum 180.41. These image statistics do not themselves establish that an input is natural.

For the fitted depth-16 full-MNIST model `105`, we minimized TV subject to zero first preactivations, a black two-pixel border, and mean ink at most 1/5. The resulting rounded eight-bit image remains a verified counterexample at $c = .001$, but its TV is 284.93, above every test image. This search therefore does *not* refute a trusted region with TV at most 150.

We can rigorously exclude this particular mechanism from that region. An exact rational dual certificate gives

$$\mathrm{TV}(p) \geq 284.87765209159$$

for every image satisfying the stated border and ink constraints and exactly zero first preactivations. More generally, the same certificate implies that an image in the border/ink region with TV at most 150 has

$$\max_i |a_{1,i}| \geq 0.1175517.$$

These claims exclude zero and uniformly near-zero first states; they do not bound the later Jacobians or prove stability on the region.

A separately declared check gives the same qualitative conclusion for the fitted depth-32 model `301`. Its exact rational dual certificate proves TV at least 287.04761391129 for a zero first state under the same border and ink constraints. TV at most 150 instead forces maximum absolute first preactivation at least .26750111067. None of the 4,097 prescribed nearest and stochastic pixel-grid roundings of the numerical minimizer entered the region. These are additional exclusions of this particular construction, not a uniform output-stability certificate or evidence that every fragile input has high TV.

For completeness, the certificate does not assume that the numerical linear-program solver is correct. In normalized coordinates $x = 2p - 1$, introduce $t_e \in [0, 2]$ with $\pm(x_i - x_j) \leq t_e$, so TV is one half of the linear objective $\sum_e t_e$. Write the resulting problem as minimizing $c^T y$ with equality constraints $Ay = b$, inequalities $By \leq d$, and coordinate box $l \leq y \leq u$. For arbitrary exact dyadic multipliers λ and $\mu \leq 0$, put $q = c - A^T\lambda - B^T\mu$. Then

$$c^T y \geq \lambda^T b + \mu^T d + \sum_j \min\{q_j l_j, q_j u_j\}.$$

We evaluate this expression using integer-rational arithmetic against the actual stored first-layer weights. It remains valid even when the proposed dual multipliers are inaccurate. If the equality constraint is violated by a first-preactivation vector a_1, the right side decreases by at most $\|\lambda\|_1 \|a_1\|_\infty$; dividing by two gives the stated near-zero exclusion. The exact coefficients and bound are saved for replay.

A complementary, separately declared stress test interpolated 256 deterministic opposite-parity pairs of correctly classified held-out MNIST images, using endpoints with TV at most 140, ink at most .195, and black two-pixel borders. We bisected toward output 0 and 3/8, rounded to eight-bit pixels, and retained all 512 rounded candidates satisfying TV at most 150, ink at most .2, and the same border condition. We attacked the 32 candidates with largest clean internal-noise gain for each target. All 64 selected inputs pass the numerical finite-radius sufficient tests at $c = .001$ and .003, for both scores. These are float64 evaluations of rigorous real inequalities, not interval proofs. The corresponding attacks found no violations; that observation is not used as a substitute for the sufficient tests.

At $c = .01$, all 64 selected inputs have numerical primary-score violations. Independently verified examples have TV 88.93 and 100.13, and mean ink .118 and .158. To test disturbances confined to valid activation states, we set one offending noise coordinate to zero in each selected witness and reverified the resulting trajectories: all hidden states remain in $[-1, 1]$, while the primary score changes by more than .1474 and the optional margin-band score changes by more than .1384, respectively. The exact pixel sums verify the region constraints. Figure 4 shows one such interpolation. Therefore a TV/ink restriction excludes the zero-state counterexample mechanism and improves this stress test at the smaller radii, but does not ensure the larger-radius condition. No conclusion of uniform stability on the trusted region follows from these finite samples.

P Finite-radius certificates using reachable activation states

A product of global operator norms ignores saturation. The following certificate uses activation ranges reachable from one clean input under *all* disturbances in the prescribed ball. It applies at any radius; small radii are not needed for its logical validity. The resulting bounds can become conservative or vacuous at large radii.

Write the hidden network in column-vector notation as $h_s = \tanh(a_s)$, $a_s = A_s h_{s-1}$, $1 \le s \le H$, with a scalar readout $z = v^\top h_H$. The input and its bias coordinate are fixed. Add postactivation disturbances η_s and output disturbance ξ, with

$$\sum_{s=1}^{H} \|\eta_s\|_2^2/N + \xi^2 \le r^2.$$

All clean activations, preactivations, and derivative matrices $D_s = \operatorname{diag}(\operatorname{sech}^2 a_s)$ below are fixed by this input. Set $J_s = D_s A_s$ for $s \ge 2$.

Forward enclosures. We construct a scalar B_s and a nonnegative vector C_s satisfying, simultaneously for every allowed disturbance,

$$\|\Delta h_s\|_2 \le B_s, \qquad |\Delta h_s| \le C_s.$$

Initialize $B_1 = \sqrt{N} r$, $C_1 = \sqrt{N} r\,\mathbf{1}$, $q_1^{\mathrm{lin}} = q_1^{\mathrm{tube}} = 1$, and $e_1 = 0$. Given the preceding enclosures, set componentwise

$$b_{s,i} = \min\{\|(A_s)_{i,:}\|_2 B_{s-1}, (|A_s| C_{s-1})_i\}.$$

Then the perturbed preactivation lies in $[a_{s,i} - b_{s,i}, a_{s,i} + b_{s,i}]$. For the scalar remainder

$$R_a(u) = \tanh(a+u) - \tanh(a) - \operatorname{sech}^2(a) u,$$

compute bounds $\rho_{s,i}^- \le R_{a_{s,i}}(u) \le \rho_{s,i}^+$ for $|u| \le b_{s,i}$. Its exact extrema occur among the endpoints, $u = 0$, and $u = -2a_{s,i}$ when this point belongs to the interval: the derivative vanishes exactly when $a + u = \pm a$. These extrema may be intersected with any valid quadratic curvature bound. Put $\rho_{s,i} = \max\{|\rho_{s,i}^-|, |\rho_{s,i}^+|\}$.

Choose upper bounds $\ell_s \ge \|J_s\|_{\mathrm{op}}$ and

$$L_s \ge \|U_s A_s\|_{\mathrm{op}}, \qquad U_s = \operatorname{diag}\Big(\operatorname{sech}^2((|a_{s,i}| - b_{s,i})_+)\Big)\,.$$

Update

$$q_s^{\mathrm{lin}} = 1 + \ell_s^2 q_{s-1}^{\mathrm{lin}}, \qquad e_s = \ell_s e_{s-1} + \|\rho_s\|_2,$$
$$q_s^{\mathrm{tube}} = 1 + L_s^2 q_{s-1}^{\mathrm{tube}}, \qquad B_s = \min\{\sqrt{N} r \sqrt{q_s^{\mathrm{lin}}} + e_s, \sqrt{N} r \sqrt{q_s^{\mathrm{tube}}}\}.$$

Finally set

$$C_{s,i} = \min\left\{B_s, \sqrt{N} r + \max_{u \in \{-b_{s,i}, b_{s,i}\}} |\tanh(a_{s,i} + u) - h_{s,i}|\right\}.$$

These intersections combine independent valid enclosures; they do not assume that every disturbance can attain all their endpoints together.

Why the forward bounds hold. The exact perturbed recurrence is $\Delta h_s = J_s \Delta h_{s-1} + \eta_s + R_s$, where R_s lies in the specified componentwise remainder interval. The sum of the clean linear responses to all η_j has norm at most $\sqrt{N} r \sqrt{q_s^{\mathrm{lin}}}$, by Cauchy–Schwarz over the layerwise noise norms and the products of the ℓ_j. The propagated remainder has norm at most e_s. Independently, every mean-value derivative on the reachable interval has coordinatewise absolute value bounded by U_s. Its operator norm is therefore at most $\|U_s A_s\|_{\mathrm{op}}$: writing its diagonal derivative as a diagonal contraction times U_s proves this assertion. Telescoping the nonlinear recurrence and using Cauchy–Schwarz now gives $\sqrt{N} r \sqrt{q_s^{\mathrm{tube}}}$. The coordinate bound follows by monotonicity of tanh and $|\eta_{s,i}| \le \sqrt{N} r$. These arguments close the induction, including the use of the previously intersected scalar and coordinate bounds to construct each new tube.

An output-specific interval. Define clean reverse adjoints

$$g_H = v, \qquad g_{s-1} = J_s^\top g_s, \qquad G = \sqrt{1 + N \sum_{s=1}^{H} \|g_s\|_2^2}.$$

Set $\rho_1^- = \rho_1^+ = 0$ and

$$E^- = \sum_{s,i} \min\{g_{s,i}\rho_{s,i}^-, g_{s,i}\rho_{s,i}^+\}, \qquad E^+ = \sum_{s,i} \max\{g_{s,i}\rho_{s,i}^-, g_{s,i}\rho_{s,i}^+\}.$$

Then every allowed disturbed output belongs to

$$[z - rG + E^-,\ z + rG + E^+]. \tag{22}$$

Indeed the exact output difference is $\xi + \sum_s g_s^\top \eta_s + \sum_s g_s^\top R_s$. The first two terms have absolute value at most rG; the last terms lie between E^- and E^+. One may further intersect this interval with

$$[z - T, z + T], \qquad T = r + \min\{\|v\|_2 B_H,\ |v|^\top C_H\}.$$

In particular these are finite-radius statements, not merely clean Jacobian estimates.

For the clipped score $\psi_a(z) = \max\{-a, \min\{a, z\}\}$ and any verified output interval $[l, u]$, the sufficient stability test is

$$\max\{|\psi_a(l) - \psi_a(z)|, |\psi_a(u) - \psi_a(z)|\} \le 1/16.$$

The main empirical convention uses $a = 1/2$, with $a = 1$ reported separately. A saturated clean score may satisfy this test even when the raw output interval is much wider than $1/8$. The certificate never requires a task label or assumes that the clean prediction is correct.

Computing the clean adjoints costs $O(HN^2 + Nd)$; computing all local spectral norms costs $O(HN^3 + Nd)$ for each input and radius. The construction does not store the earlier N by HN transfer arrays. Our implementation first tries cheaper global spectral bounds and reuses any certificate at a larger radius for smaller radii. Ordinary float64 evaluations of these real inequalities are reported as numerical sufficient tests; selected claims receive separate interval or exact-rational verification. Checking many data points or random cube points does not constitute a global certificate. On a small finite cube, exhaustive coverage does.

A stronger refinement using signed clean transfers. When the preceding inexpensive bounds are inconclusive, one may retain the full clean transfer matrices from all previous layers. Let T_s map RMS-normalized past disturbances to the current clean linear preactivation response, so each injected hidden-noise block initially has coefficient $\sqrt{N}I$. If all earlier nonlinear remainders have stacked component bounds $\rho^- \le R \le \rho^+$, then each current preactivation displacement lies in

$$\left[-r\|(T_s)_{i,:}\|_2 + \frac{((T_s)_+\rho^- + (T_s)_-\rho^+)_i}{\sqrt{N}},\quad r\|(T_s)_{i,:}\|_2 + \frac{((T_s)_+\rho^+ + (T_s)_-\rho^-)_i}{\sqrt{N}}\right].$$

Here $(T_s)_+$ and $(T_s)_-$ denote its positive and negative parts. The exact scalar remainder extrema above apply on this asymmetric interval as well. Update the transfer matrix with the clean derivative and append the new $\sqrt{N}I$ noise block. At the output use the same signed interval propagation and include scalar output noise in the linear Euclidean coefficient norm. This is an exact algebraic decomposition of the perturbed recurrence, followed by interval bounds; no linearization remainder has been discarded. It strengthens the earlier recurrence that propagated only absolute remainder bounds. Its cost is $O(H^2N^3)$, so it is used only where the cheaper certificates are inconclusive.

A verified saturated example. The preceding sufficient inequalities are mathematical statements; the large empirical tables evaluate them in ordinary float64 and are not themselves outward-rounded interval proofs. We separately verified one illustrative finite-radius saturation certificate using outward intervals and exact integer-rational matrix bounds. For the depth-32, width-128 MNIST 3-versus-8 SGD model used in the finite-grid counterexample, the fourth input in the fixed uniform held-out sample has clean output in a narrow interval around 8.282734267956. At $c = 1/16$, reachable-tube derivative bounds prove

$$|z(x;\eta,\xi) - z(x;0,0)| < 6.559413$$

for every simultaneous hidden/output disturbance in the prescribed ball. Consequently the clipped output is unchanged, for both cap $1/2$ and cap 1. This is a pointwise, all-disturbances certificate on one actual held-out image; it is not a claim about every input. The same saved network has the verified finite-grid global counterexamples at $c = 10^{-3}$ described above. Thus the distinction between stability on data and global stability is present even in independently verified examples, and clipping can certify a useful condition when the raw-output variation bound is much larger than $1/16$.

For this computation, the clean trajectory is enclosed with 90-decimal outward interval arithmetic. At each layer we round the reachable-tube upper derivative bounds upward to exact dyadic numbers. For the resulting matrix $A = W\,\mathrm{diag}(D)$, the stored floating-point SVD merely proposes dyadic factors U, S, V^T. All Gram matrices and the residual are recomputed with exact integers. Writing

$$a = \|U^TU\|_\infty,\quad b = \|V^TV\|_\infty,\quad t = \max_i |S_{ii}|,\quad e^2 = \|A - USV^T\|_F^2,$$

we use the rational inequality

$$\|A\|_{\mathrm{op}}^2 \le (1+\epsilon)t^2ab + (1+\epsilon^{-1})e^2,\qquad \epsilon > 0.$$

Here the matrix infinity norm is the maximum absolute row sum. The bound follows from the symmetric Gram-matrix spectral bound, the triangle inequality, and Young's inequality, and makes no accuracy assumption about the proposed SVD. Product rounding when constructing A is enclosed by an additional exact Frobenius residual. This yields a replayable certificate with saved dyadic factors. It still trusts the interval library for transcendental evaluations and is not a Lean formalization.

A covariance refinement for a difficult vertex. Products of clean Jacobian norms can also be replaced by the exact covariance of the clean linear response. In column-vector notation, write $J_s = D_s A_s$ for the clean hidden-state Jacobian, set $G_1 = I$, and define

$$G_s = I + J_s G_{s-1} J_s^{\mathsf{T}} \qquad (s \geq 2).$$

If T_s maps the stacked hidden disturbances to the clean linearized state, then $G_s = T_s T_s^{\mathsf{T}}$ by induction. This is a deterministic Gramian identity; the certificate makes no Gaussian assumption about the disturbances. Thus its linear part has norm at most $\sqrt{N} r \sqrt{\|G_s\|_{\mathrm{op}}}$. Suppose coordinatewise Taylor remainder intervals on the already established preactivation tube imply a vector norm bound R_s. With $\rho_1 = 0$, the recursion

$$\rho_s = \|J_s\|_{\mathrm{op}} \rho_{s-1} + R_s, \qquad B_s = \sqrt{N} r \sqrt{\|G_s\|_{\mathrm{op}}} + \rho_s$$

then bounds the nonlinear remainder and total state displacement, respectively. Indeed, subtracting the exact clean linear response gives $e_s = J_s e_{s-1} + R_s(\Delta a_s)$, while each preactivation coordinate lies within $\|A_{s,j:}\|_2 B_{s-1}$ of its clean value. Finally,

$$|z^\eta - z| \leq \|v\|_2 B_H + r$$

is a valid output bound, including the simultaneous scalar disturbance. Using the full radius in both terms only makes this last bound conservative.

The implementation encloses each G_s by outward-rounded dyadic interval matrix products. An arbitrary dyadic midpoint proposal M gives $\|G_s\|_{\mathrm{op}} \leq \|M\|_{\mathrm{op}} + \sup_{G \in [G_s]} \|G - M\|_F$; both terms are bounded with exact rational matrix arithmetic and outward square roots. Scalar remainder extrema occur at the two tube endpoints, zero, and $-2a$ when that critical point lies in the tube; intersecting this enclosure with $|R(u)| \leq u^2/2$ is valid because $|\tanh''| \leq 1$. No floating estimate of a covariance eigenvalue is accepted as a certificate.

Exact clean-output enclosures at dataset scale. For the finite held-out set, the stored float32 inputs and weights are dyadic rational numbers. We approximate hidden states at denominator $Q_h = 2^{24}$ and weights at denominator $Q_w = 2^{30}$, and compute all approximate affine maps by integer matrix multiplication. Before every product, an exact bound on the sum of absolute products rules out signed 64-bit overflow, including intermediate accumulation. The resulting affine arguments are therefore exact rational numbers.

The activation computation uses a piecewise-linear tanh table on $[-12, 12]$ with grid spacing $h = 1/4096$. Every proposed table value is independently enclosed using outward interval arithmetic. Since $|\tanh''| \leq 1$, interpolation adds at most $h^2/8$ to the maximum table-entry error; rounding the interpolated value adds at most $1/(2Q_h)$. Outside the table, the approximation is the appropriate sign and its error is at most $2e^{-24}$. Thus every approximate activation has a common rigorously established coordinate error τ, without trusting floating-point evaluations of tanh.

If a clean hidden state is enclosed around its approximate value with Euclidean error e, then the next state's error is at most

$$e' \leq \|W\|_{\mathrm{op}} e + \sqrt{N}\, \|W - \widetilde{W}\|_F + \sqrt{N}\, \tau.$$

Here $\|W\|_{\mathrm{op}}$ has an exact matrix-factor certificate, the quantization residual is evaluated exactly, and the approximate hidden coordinates remain in $[-1, 1]$. The first layer separately accounts for input quantization through the exact identity

$$xW - \widetilde{x}\widetilde{W} = x(W - \widetilde{W}) + (x - \widetilde{x})\widetilde{W}.$$

It uses column sums for the first term and exact integer products for the second. The final scalar error is at most $\|v\|_2 e + \|v - \widetilde{v}\|_1$. These bounds produce a clean-output interval at every recorded input. Combining each interval with an independently verified global disturbance bound gives a pointwise clipped-score certificate whenever the resulting interval comparison succeeds. Every remaining input must receive a separate local proof before a full held-out-set conclusion is drawn. The method certifies the specified finite dataset, not the input distribution or all possible images.

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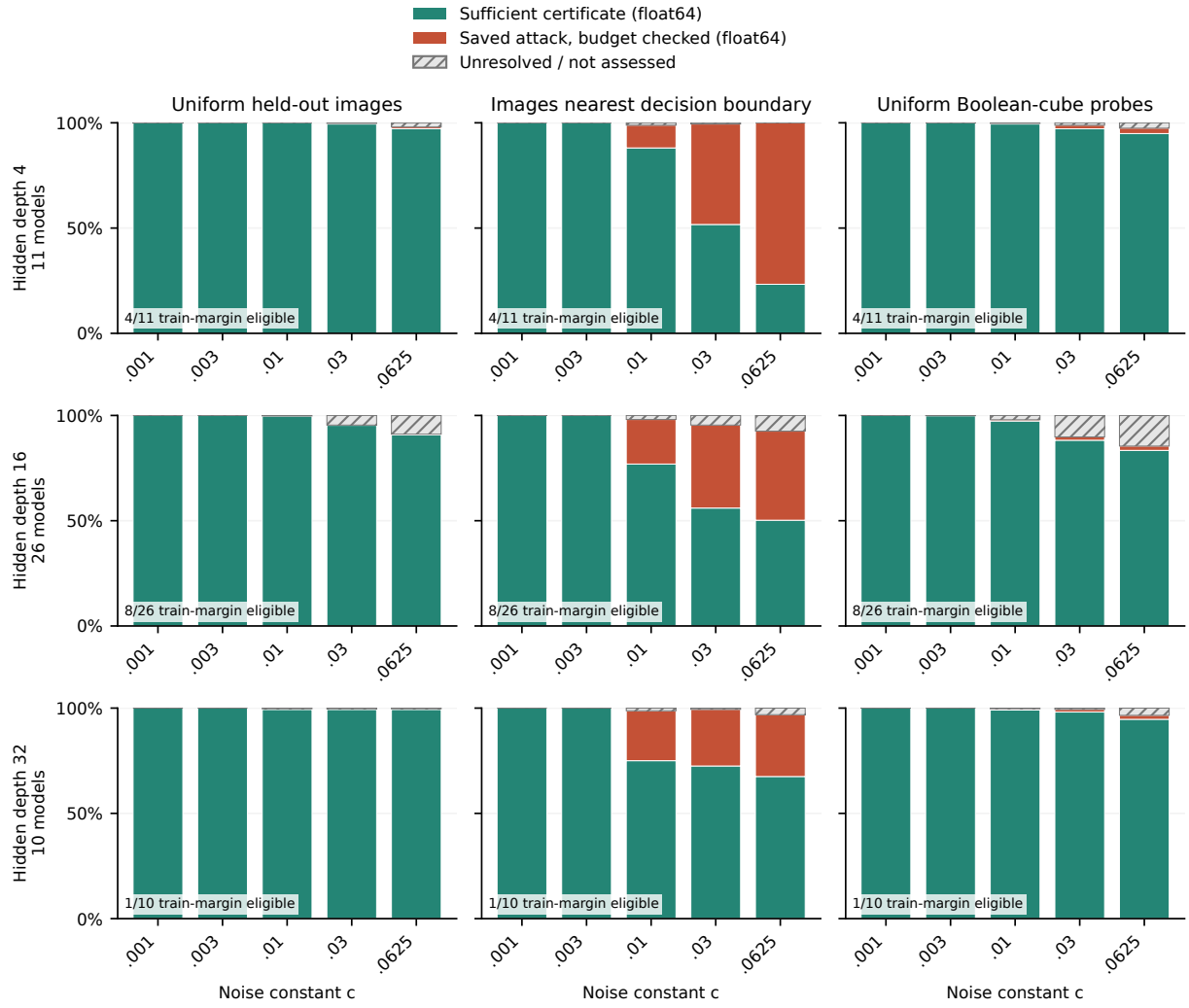


Figure 1: Fixed-subset audit of all depth-4/16/32 unconstrained tanh runs, including failed training fits. Green denotes a sufficient finite-radius certificate evaluated in float64; red denotes an explicit saved attack that passes a float64 budget and score recheck; hatched gray denotes an unresolved or unassessed point. Selected exact verifications are reported separately. The same points and denominators are used at every radius. Boundary images and uniform Boolean probes have different sampling interpretations from uniformly selected test images.

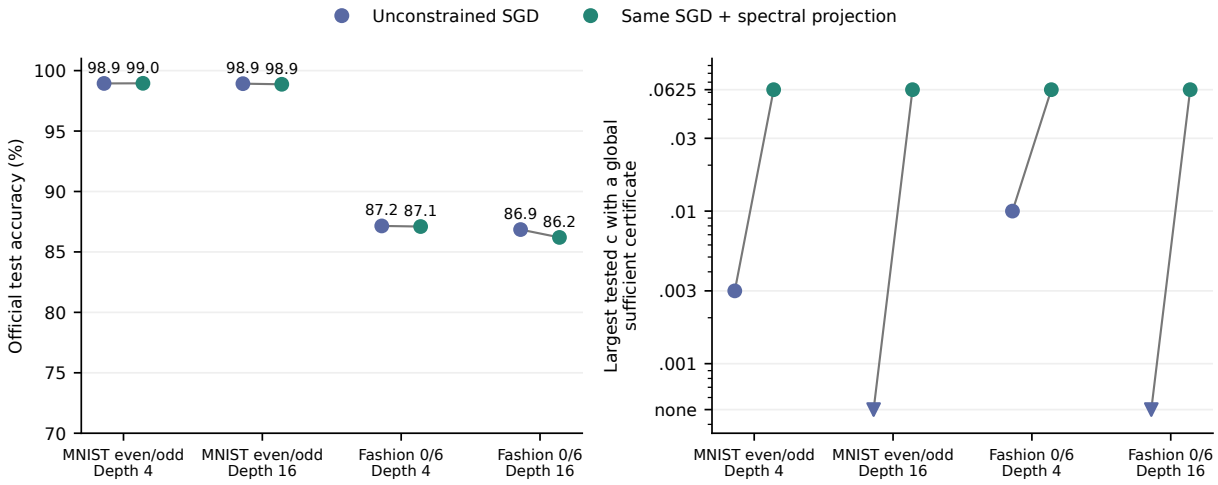


Figure 2: Matched vanilla-SGD comparison with and without a spectral projection after each epoch. The right panel gives the largest tested radius constant with a global sufficient certificate, a lower bound on stability rather than an estimate of the largest valid radius. Training-margin eligibility is reported separately from accuracy.

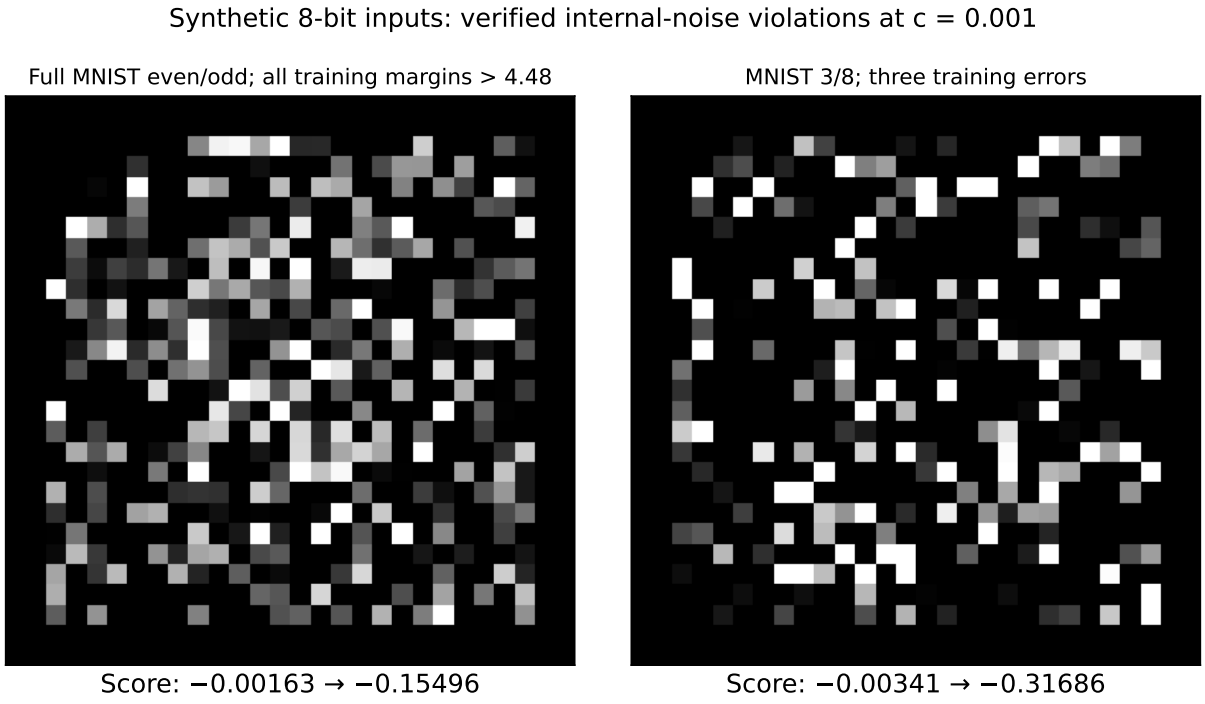


Figure 3: Synthetic, valid eight-bit inputs violating the global clipped-score condition at $c = .001$. Both have a fixed black border and low total ink. The left network was trained on the full MNIST even-versus-odd task and has exactly verified training margins above 4.4815762; the right network was trained on digits 3 and 8. The images themselves are unchanged by the internal disturbances. These are off-data counterexamples, not mislabeled examples from a test set.

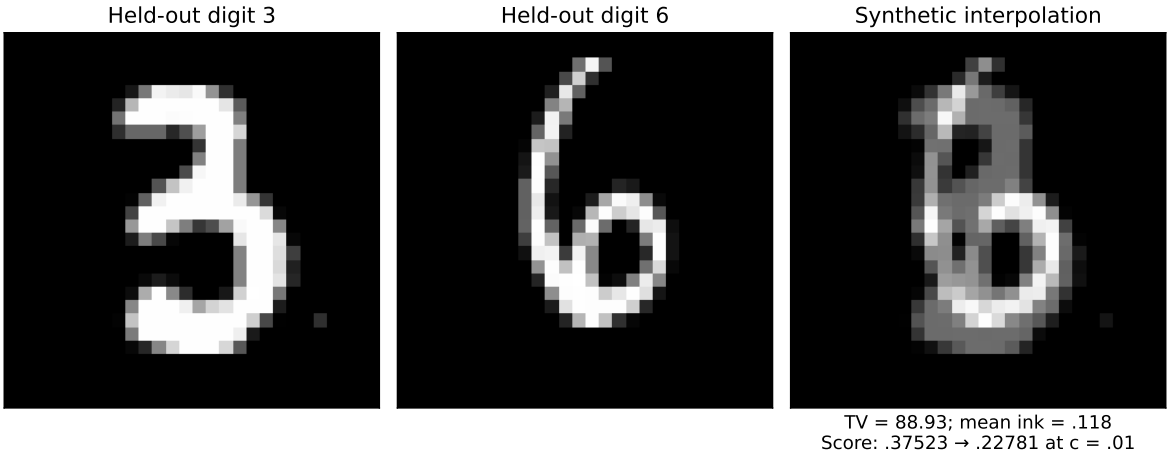


Figure 4: An interpolation of opposite-parity held-out images, rounded to the eight-bit grid. The interpolated image is synthetic and is not assigned a parity label. At $c = .01$, a verified internal disturbance changes its clipped score by more than .1474, while every hidden state stays in $[-1, 1]$. Its TV and mean ink are within the stated smooth-image region. This is a larger-radius example; the selected interpolation set passes numerical sufficient tests at $c = .001$ and .003.